A CERTIFIED SIMPLE, EVEN GROUND STATE OF THE LOCALIZED WEIL QUADRATIC FORM FOR ALL WINDOWS $L \in (0, 2]$

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Abstract. Let Q_L be Weil's quadratic form for the Riemann zeta function, restricted to real test functions supported in $[-L, L]$ on the logarithmic scale; it contains the pole terms, the archimedean term and every prime power n with $\log n < 2L$. Connes and van Suijlekom proved that, for a quadratic form of this type that defines an essentially self-adjoint operator on trigonometric polynomials, a simple lowest eigenvalue with even eigenfunction ξ forces all zeros of $\hat{\xi}$ to be real; we do not verify that domain hypothesis for our operator and draw no conclusion about zeros. For the Weil form the "simple-even" property is one of the missing steps in the strategy of Connes, Consani and Moscovici. We give a computer-assisted proof, in Arb ball arithmetic and exact integer arithmetic, that the ground state of Q_L is simple and even for every $L \in (0, 2]$. The proof combines min-max monotonicity in L with four pieces: a Lipschitz argument on $(0, 0.05]$; a covering of $[0.05, 1.19]$ by 31 certified windows based on Zhu's frequency-cutoff reduction; a covering of $[1.19, 1.4035]$ by 9 windows in which second-eigenvalue lower bounds come from Elston's window-compressed prime-norm certifier combined with a rank-one deflation; and a covering of $[1.4035, 2]$ by 54 windows with value certificates $\lambda \geq \mu$ from a full-source Schur elimination. The last piece follows the positivity method of two AI-agent preprints on rStation (bramble_trace; lihongmin), which we re-derived and re-implemented in a clean room and extended by a spectral shift, a rank-one deflation and an exact-integer structured LDL^{T} factorisation. Upper bounds come from Legendre–Galerkin Rayleigh quotients with rigorous quadrature bounds. The pre-registered grid of the last piece failed at one window. We close that gap in two ways: by a parameter rule registered after the failure, which we report as post hoc, and, without that rule, by monotonicity from a registered certificate at the next window. Positivity on this range is not new. On $(0, 0.8]$ the simple-even statement had been claimed before. On $(0.8, 2]$ we know of no certified proof of the simple-even statement or of second-eigenvalue lower bounds; an unrefereed claim for all windows, which we do not regard as established, is discussed in §1.3. As by-products we give rigorous upper bounds for the lowest eigenvalue in both parity sectors up to $L = 2.25$. The work is not refereed; all reviews so far were carried out by AI agents, from two AI systems.

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1. Introduction

1.1. **The localized Weil form.** For a real function $f \in C_c^\infty(-L,L)$ put $\tilde f(x) = f(-x)$, $g = f * \tilde f$ and $F(z) = \hat f(z) = \int f(x)e^{izx}\,dx$. Weil's explicit-formula functional [36], evaluated at g, gives the quadratic form

$$Q_L(f) = W(g), \qquad W(g) = \hat g(\tfrac{i}{2}) + \hat g(-\tfrac{i}{2}) - g(0)\log\pi + \frac{1}{2\pi}\int_{\mathbb R} \hat g(r)\,\operatorname{Re}\psi\big(\tfrac14 + \tfrac{ir}{2}\big)\,dr - \sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt n}\big[g(\log n) + g(-\log n)\big], \tag{1.1}$$

where $\psi = \Gamma'/\Gamma$ and Λ is the von Mangoldt function. Weil's criterion states that the Riemann hypothesis (RH) is equivalent to $Q_L(f) \ge 0$ for all such f and all $L > 0$ [36, 4]. In §2 the form is extended by closure to a form domain in $L^2(-L,L)$. Since g is supported in $[-2L,2L]$, only prime powers with $\log n < 2L$ contribute; the prime power n enters at $L = \frac12\log n$ (Table 1).

Table 1. Prime powers n and the window $L = \frac12\log n$ at which they enter the form, for $L \le 2$.

n	2	3	4	5	7	8
$\frac12\log n$	0.3466	0.5493	0.6931	0.8047	0.9730	1.0397
n	9	11	13	16	17	19
$\frac12\log n$	1.0986	1.1989	1.2825	1.3863	1.4166	1.4722
n	23	25	27	29	31	32
$\frac12\log n$	1.5677	1.6094	1.6479	1.6836	1.7170	1.7329
n	37	41	43	47	49	53
$\frac12\log n$	1.8055	1.8568	1.8806	1.9251	1.9459	1.9851

1.2. **The simple-even hypothesis.** Connes and van Suijlekom [13, Thm. 6.1] prove: let D be a real distribution on an interval and let Q be the quadratic form with Schwartz kernel $\tilde D(x-y)$, defined on trigonometric polynomials; if Q defines a lower-bounded *essentially self-adjoint* operator (so that the trigonometric polynomials are a core) whose lowest spectral value is a *simple, isolated* eigenvalue with *even* eigenfunction ξ, then all zeros of the entire function $\hat\xi$ are real. (Their introduction states the result without the essential self-adjointness hypothesis and refers to Theorem 6.1 for the precise form.) Our operator A_L is instead the self-adjoint operator of the closure of Q_L on $C_c^\infty(-L,L)$ (§2). We have not proved that trigonometric polynomials form an operator core for A_L, nor any form-core substitute for that hypothesis, so *we do not claim that the Fourier transform of our ground state has only real zeros*; our theorem is a statement about A_L only. Connes, Consani and Moscovici [12] propose a strategy towards RH based on a spectral realization of the zeros through the eigenfunctions ξ_λ of the

Weil form on growing windows; in [12, §8] they list two "missing steps": (i) the smallest eigenvalue of the Weil quadratic form QW_λ is simple with even eigenvector, and (ii) the zeros of $\hat{\xi}_\lambda$ converge to the nontrivial zeros of ζ. The same two steps are listed in [11, §6.6]. The present paper addresses only step (i), and only on the bounded range $L \in (0, 2]$; it says nothing about step (ii) or about large L, and does not prove positivity beyond what the certified lower bounds imply on this range. In the conventions of [13] the operator acts on $L^2([-\frac{L}{2}, \frac{L}{2}])$, so their L is twice ours; in [12] the window is $[\lambda^{-1}, \lambda]$ on the multiplicative scale, i.e. $L = \log \lambda$. We claim agreement of our form with theirs only up to a positive affine rescaling (Remark 2.4, §11). The second AI review (§10.3) matched the pole, prime and archimedean terms of (1.1) against [12, (3.7)–(3.11)] under $u = e^x$, with their $2\theta'(t)$ equal to our $H(t)$ of §5.4; no human has checked this.

1.3. **Prior work.** We list what we know to have been proved or claimed; status labels follow the sources. Many items are unrefereed 2026 deposits or preprints, some AI-assisted; papers on rStation are accepted by AI reviewer agents and are not refereed. Our literature search (arXiv, Zenodo, GitHub and rStation) was last refreshed on 31 October 2026; the items [30, 10], which the second AI review found, were added on 1 November 2026.

- *Small windows.* Suzuki [33, Thm. 1.4] proves for the full form that the lowest eigenvalue is positive, simple and has an even eigenfunction for $L \in (0, a_0)$ with a_0 not explicit (Kato perturbation theory). The mathia notes PL-294 [25] (unrefereed) show that the full form satisfies the Beurling–Deny criterion iff $L \le \frac{1}{2} \log \rho_{\text{plastic}} \approx 0.1406$, which gives a simple ground state of fixed sign. Andrade [2] (unrefereed) proves a Perron-type theorem for the pole-removed form and states the same threshold for the full kernel in a remark. An analytic argument for $L \le 9/50$ is reported in [18]; we have not verified it.
- *Zhu* [38] introduced the frequency-cutoff ("one-stroke") reduction with a head/tail block estimate [Zhu, Thm. 1.1] on which all our window certificates rest, certified simple-even at the single window $L = 0.8$ [Zhu, Thm. 6.2], and certified variational upper bounds on the lowest eigenvalue for $L \in [0.5, 2]$ [Zhu, Table 3]. Zhu [§7] retracts an earlier positivity claim at $L = 1.19$.
- *Kramarenko-Byrd* [22] (unrefereed) claims a rigorous proof of simple-even for all $0 < L \le 4/5$ in our half-width convention. We have not checked it. If it is correct, our result on $(0, 0.8]$ is an independent confirmation.
- *A claim for all windows.* Siche [30] (GitHub repository, first commit 3 April 2026, last change 7 April 2026; unrefereed) claims that the smallest eigenvalue of QW_λ is simple with even eigenvector for every window, through Krein–Rutman positivity of the prime-sum operator and an $O(\log \lambda)$ bound for the even pole term. We have not checked it in full, and we do not regard it as an established proof: we could not reconcile two of its steps, both raised by the second AI review (§10.3). A finite sum of compressed translations is in general not compact, and in our normalisation the even pole term $2F(\frac{i}{2})^2 = 2\langle f, \cosh(x/2)\rangle^2$ has norm $2\int_{-L}^{L} \cosh^2(x/2)\,dx = 2(L + \sinh L)$, which grows like $\lambda = e^L$, not like $\log \lambda$. If that claim is correct, it contains our theorem.
- *Perturbative simplicity* on a neighbourhood of $L = 0.8$ of non-explicit radius: PL-287 [25].
- *Positivity and per-sector lower bounds* (no simplicity or parity statement): jose-paul-b [21] for $L \le \frac{1}{2} \log 7$; Liu [24] at $L = 17/16$ (block-Schur bounded comparison, $Q \ge 2^{-49162}$); Mori [26] at $L = \frac{1}{2} \log 15$; Senke [29] at $L = 77/64$; Elder [15] at $L = 0.8$; and Elston [16], whose certified lowest-eigenvalue lower bounds at $L = 1.19$, 1.3 and

1.495 (e.g. $\lambda \geq 3.501 \cdot 10^{-96}$ at $L = 1.495$) use the certifier we build on in Piece III. On rStation, lihongmin [23, Thm. 1.1] certifies strict positivity of both parity sectors for every support radius $0 < a \leq \frac{1}{2}\log 11$, and bramble_trace [8, Thm. 1.1] for every $0 < a \leq 2$, both by block Schur elimination with the complete archimedean response. Their form [23, (2.1)], [8, (2.1)] coincides term by term with (1.1), with $a = L$. Neither paper identifies the ground state or compares the two sectors ([23, §1]: "our lower certificate neither identifies the ground state nor estimates the sharp minimum"), and [8] gives no explicit eigenvalue lower bound. Piece IV of this paper (§7) is built on the method of [8], with the Green-response identity of [23]; our clean-room implementation independently re-checks their positivity statements in the odd sector at each of our 53 grid windows in $(1.4035, 2]$ (every odd certificate has $\mu > 0$) and in both sectors at $a = 2$ (a pilot run with $\mu = 0$; its records are not in this release and are available on request), hence by Lemma 3.1 on $(0, 2]$, non-strictly in the even sector. An unreviewed forum follow-up to [8] reports values that order the two parity sectors at the single window $a = 2$.

- *Upper bounds for large L.* Without hypotheses on the zeros, Mori [27, Thm. K(a)] proves $\lambda_{\min} \leq 3.50 \times 10^{24}\mu^{9/2}\log\mu\, e^{-4\pi\mu}$ for $50 \leq \mu = e^{2L} \leq 10^4$, in his normalisation, and bramble_trace [9, Thm. 2.1] proves $\lambda_1^{\mathrm{ev}}(L) \leq 10^{16}\exp(11L - 2\pi e^{2L})$ for $L \geq 1$ in the same form as ours. At the sharper scale and also without hypotheses on the zeros, bramble_trace and span_of_dilates [10, Thm. 1, Cor. 2] (published 31 October 2026, 19:33 UTC) prove absolute sampling bounds and, for each fixed index n, upper bounds $\lambda_n(L) \leq \exp(-4\pi d + C_n(1 + L)^2)$ in each parity sector for $L \geq L_n$, in their setting of Dirichlet characters with $d = e^{2L}/q$; the constants C_n, L_n are not explicit, and there is no lower bound or simplicity statement. Under RH, span_of_dilates [32, Thm. 1] proves $-\ln\lambda^*(L) \geq 4\pi e^{2L} - O(L^2)$ in both sectors, in Zhu's normalisation. The decay scale $e^{-4\pi e^{2L}}$ also appears in Connes's prolate heuristic [11] and in the model and conjecture of [31, Prop. 4.1, Conj. 4.2(a)] (stated there for real odd characters).
- *Finite matrices.* Andrade [1] and Andrews Jr. [3] certify simple-even for finite Galerkin matrices QW^N; these are not statements about the continuum form.
- *Classical ancestor.* Yoshida [37] combined finitely many pivots with a rigorous high-frequency complement lemma.

1.4. **Contributions.**

(1) A computer-assisted proof that the ground state of Q_L is simple and even for every $L \in (0, 2]$ (Theorem 4.1). The part we believe to be new is $(0.8, 2]$ (literature search including rStation, 31 October and 1 November 2026): there, a certified proof that the ground state is simple and even, and the certified lower bounds for λ_2^{ev} that prove it. An unrefereed claim of simplicity for all windows [30], which we do not regard as established, is discussed in §1.3. Positivity on this range is not new [23, 8]. The piece $(0, 1.19]$ was released earlier as an unrefereed note [5] and $(1.19, 1.4035]$ in release 1.1 of this paper [6]; this version reproduces their mathematical content and adds $[1.4035, 2]$.

(2) A rank-one deflation (Lemma 5.14), which turns a certified lowest-eigenvalue bound into a certified *second* even eigenvalue bound: added to Elston's certifier in Piece III (Corollary 5.15) at no extra assembly cost, and lifted to the operator in Piece IV, where it raises the displacement rank of the core Schur complement from 2 to 4. The rStation screen of 31 October 2026 found no bound on λ_2^{ev}; [8] proves positivity of both sectors up to $a = 2$ without explicit eigenvalue lower bounds.

(3) A clean-room, general-L re-derivation and re-implementation of the full-source Schur certificate of [8] as a *value* certificate $\lambda_k^\sigma(L) \geq \mu$, with an exact-integer structured LDL^T factorisation (Lemma 7.4) in place of a floating-point one, independent per-run gates, and a pre-registered grid whose one failure and post-hoc repair are reported in full (§7).
(4) Rigorous upper bounds for the lowest even and odd eigenvalues up to $L = 2.25$ from a fast Legendre convolution recurrence with a rigorous Gauss–Legendre bound (§8). We claim only explicit rigorous values at $L = 2.25$, the sharpest we know of there; analytic upper bounds with double-exponential decay are due to Mori [27, Thm. K] and bramble_trace [9, Thm. 2.1] (unconditional, explicit), to bramble_trace and span_of_dilates [10, Cor. 2] (unconditional, at the $4\pi e^{2L}$ scale, every fixed index, non-explicit constants) and to span_of_dilates [32, Thm. 1] (under RH). No value is claimed to be close to the eigenvalue.
(5) Numerical measurements of the weak singular points of the computed minimisers at $|y| = |L - \log n|$ reported by Eberwein [14, §13] (§9); their location is not new.

Not claimed as new: the range $(0, 0.8]$ [22] and in particular $(0, 0.1406]$ [25]; the window $L = 0.8$ by itself [Zhu, Thm. 6.2]; positivity on $(0, 2]$ [23, 8]; the ordering of the two sectors at the single window $L = 2$ (forum follow-up to [8]); the full-source Schur architecture, the Green-response identity and the trigonometric bases of Piece IV [23, 8]; the $4\pi e^{2L}$ decay scale of the lowest eigenvalue [11, 27, 31, 32, 10], including unconditional upper bounds at that scale [10]; certified upper bounds on the lowest eigenvalue for $L \in [0.5, 2]$ [Zhu, Table 3]; the frequency-cutoff reduction and block estimate [Zhu, Thm. 1.1]; the window-compressed prime-norm reduction and its implementation [16]; the Legendre convolution recurrence [19], already used for this problem by Liu [24].

2. Setup and normalisation

2.1. **Frequency side.** For real f supported in $[-L, L]$ one has $\hat{g}(t) = F(t)F(-t) = |F(t)|^2$ for real t, and by Plancherel $g(0) = \|f\|^2 = \frac{1}{\pi}\int_0^\infty |F|^2$. Hence

$$Q_L(f) = [\text{pole}] + \frac{1}{\pi}\int_0^\infty \Psi_L(t)|F(t)|^2\,dt, \tag{2.1}$$

$$\Psi_L(t) = \operatorname{Re}\psi\big(\tfrac{1}{4} + \tfrac{it}{2}\big) - \log\pi - \sum_{\log n < 2L} \frac{2\Lambda(n)}{\sqrt{n}}\cos(t\log n),$$

with $[\text{pole}] = 2F(\frac{i}{2})F(-\frac{i}{2})$. Here $F(\frac{i}{2}) = \int f e^{-x/2}$ is real; for even f, $[\text{pole}] = +2F(\frac{i}{2})^2$, and for odd f, $[\text{pole}] = -2F(\frac{i}{2})^2$. This normalisation is that of Zhu [Zhu, §6]; it was checked numerically against the zero side of the explicit formula (first 400 zeros).

2.2. **Position side.** Let γ denote Euler's constant. For $g = f * \tilde{f}$,

$$\begin{aligned} Q_L(f) = 4\int_0^{2L} g(x)\cosh\tfrac{x}{2}\,dx - g(0)\big(\log\pi + \gamma + \log(1 - e^{-4L})\big) \\ + \int_0^{2L} \frac{2\,[e^{-2x}g(0) - e^{-x/2}g(x)]}{1 - e^{-2x}}\,dx - 2\sum_{\log n < 2L}\frac{\Lambda(n)}{\sqrt{n}}\,g(\log n). \end{aligned} \tag{2.2}$$

Derivation. For smooth f, insert $\psi(s) = -\gamma + \int_0^\infty \frac{e^{-t} - e^{-st}}{1 - e^{-t}}\,dt$ into (1.1) and apply Fourier inversion (Fubini applies since g is smooth). For $x > 2L$ only the $g(0)$ part of the integrand survives and integrates in closed form, giving the term $\log(1 - e^{-4L})$; the pole term gives

$4\int_0^{2L} g\cosh(x/2)$. For a polynomial f on $[-L,L]$ (needed for the trial functions), g is a polynomial on $[0,2L]$ and Lipschitz on $\mathbb{R}$; apply (2.2) to the mollification $f*\phi_\epsilon$ on the window $L+\epsilon$ (no $\log n$ equals $2L$, and the $\log(1-e^{-4L})$ term absorbs the change of upper limit) and let $\epsilon\to 0$: the frequency side converges by dominated convergence since $\Psi_L|F|^2 = O(t^{-2}\log t)$, and on the position side $g_\epsilon \to g$ uniformly with a uniform Lipschitz constant, which dominates the integrand $\sim (g(0)-g(x))/x$ near $x=0$.

2.3. **The operator, parity and discreteness.** *Closability.* On real $f\in C_c^\infty(-L,L)$ write $Q_L = q_H + b_L$, where $q_H(f) = \frac{1}{\pi}\int_0^\infty H(t)|F(t)|^2\,dt$ with $H(t) = \operatorname{Re}\psi(\frac14+\frac{it}{2}) - \log\pi$ (the archimedean and $\log\pi$ terms of (1.1)), and b_L collects the pole and prime terms. By Lemma 5.6 below, $H(t)\ge H(0)$. Hence q_H is the restriction to $C_c^\infty(-L,L)$ of the form $f\mapsto \frac{1}{2\pi}\int_{\mathbb{R}} H|\hat f|^2$ of a real, lower-bounded Fourier multiplier on $L^2(\mathbb{R})$, with its maximal domain $\{\int |H||\hat f|^2<\infty\}$; that form is closed, and so is its restriction to the closed subspace of functions supported in $[-L,L]$. A restriction of a closed form is closable, so q_H is closable. The form b_L is bounded on $L^2(-L,L)$: $|F(\pm\frac{i}{2})|\le \|f\|(2\sinh L)^{1/2}$ by Cauchy–Schwarz, so the pole term is at most $4\sinh L\,\|f\|^2$ in absolute value, and $|g(\pm\log n)|\le\|f\|^2$ bounds the prime term by $2\sum_{\log n<2L}\Lambda(n)n^{-1/2}\|f\|^2$. A closable lower-bounded form plus a bounded symmetric form is closable and lower bounded, with the same form domain. Let $\bar Q_L$ be the closure of Q_L on $C_c^\infty(-L,L)$ and A_L the associated self-adjoint operator on $L^2(-L,L)$; its form domain is the closure of $C_c^\infty(-L,L)$ in the norm $(\frac1\pi\int_0^\infty(1+|H|)|F|^2)^{1/2}$. We write Q_L for $\bar Q_L$ where no confusion arises.

Discreteness. By Lemma 5.4 below, $\Psi_L(t)\ge \log(t/2\pi) - 1/t - \sum_{\log n<2L}2\Lambda(n)n^{-1/2}\to\infty$, and the pole term is bounded by $C\|f\|^2$; hence the form norm controls $\int\log(2+t)|F|^2$. For functions supported in $[-L,L]$ the set where this quantity and $\|f\|$ are bounded is compact in L^2 (high frequencies are uniformly small, and on compact t-sets the F are equicontinuous by Paley–Wiener). So A_L has compact resolvent, in agreement with [12, Thm. 3.6]. The kernel is real, so we work with real f throughout.

Lemma 2.1 (Parity splitting)**.** *Let $(Rf)(x)=f(-x)$. Then $Q_L(Rf)=Q_L(f)$, and $Q_L(f_e+f_o)=Q_L(f_e)+Q_L(f_o)$ for f_e even and f_o odd. Hence $A_L = A_L^{\mathrm{ev}}\oplus A_L^{\mathrm{odd}}$.*

Proof. Rf has transform $F(-t)$, which leaves $|F(t)|$ (t real) and $F(\frac i2)F(-\frac i2)$ unchanged. For the bilinear form B of Q_L, $B(f_e,f_o)=B(Rf_e,Rf_o)=-B(f_e,f_o)$. □

Definition 2.2. For each parity sector $\sigma\in\{\mathrm{ev},\mathrm{odd}\}$ let $\lambda_1^\sigma(L)\le\lambda_2^\sigma(L)\le\cdots$ be the eigenvalues of A_L^σ, counted with multiplicity; by compactness of the resolvent they coincide with the min-max values

$$\lambda_k^\sigma(L) = \inf_{\substack{V\subset\mathcal{D}^\sigma\\ \dim V=k}}\ \sup_{0\neq f\in V}\frac{Q_L(f)}{\|f\|^2},$$

where $\mathcal{D}^\sigma$ is the form domain of the sector. (Indices start at 1, as in [38, 16]; the earlier note [5] writes $\lambda_0,\lambda_1^{\mathrm{ev}},\lambda_0^{\mathrm{odd}}$ for our $\lambda_1^{\mathrm{ev}},\lambda_2^{\mathrm{ev}},\lambda_1^{\mathrm{odd}}$.)

Corollary 2.3. *The lowest eigenvalue of A_L is simple with even eigenfunction if and only if*

$$\lambda_1^{\mathrm{ev}}(L) < \min\bigl(\lambda_2^{\mathrm{ev}}(L),\lambda_1^{\mathrm{odd}}(L)\bigr).$$

Remark 2.4. The property in Corollary 2.3 is invariant under $Q\mapsto aQ+b\|\cdot\|^2$ with $a>0$. Conventions that differ from ours by a positive factor (e.g. 2π) or a constant shift (e.g. the placement of $\log\pi$) therefore do not affect the theorem.

3. Monotonicity and the grid argument

Lemma 3.1 (Monotonicity in L)**.** *For $0 < L < L'$, every $k \geq 1$ and $\sigma \in \{\mathrm{ev}, \mathrm{odd}\}$, $\lambda_k^\sigma(L') \leq \lambda_k^\sigma(L)$.*

Proof. Extension by zero $\iota : L^2(-L, L) \to L^2(-L', L')$ is an isometry that preserves parity and maps the form domain into the form domain, and $Q_{L'}(\iota f) = Q_L(f)$: the pole and archimedean terms depend only on F, and a prime power with $2L \leq \log n < 2L'$ contributes $g(\log n) = 0$ because g is continuous and supported in $[-2L, 2L]$ with $g(\pm 2L) = 0$. The min-max infimum over the larger family of subspaces of $L^2(-L', L')$ (which contains all ιV) can only be smaller. □

Corollary 3.2 (Step criterion)**.** *Let $L_i < L_{i+1}$, let $U(L_i) \geq \lambda_1^{\mathrm{ev}}(L_i)$, and let $\ell_2(L_{i+1}) \leq \lambda_2^{\mathrm{ev}}(L_{i+1})$ and $\ell^{\mathrm{odd}}(L_{i+1}) \leq \lambda_1^{\mathrm{odd}}(L_{i+1})$. If*

$$\min\big(\ell_2(L_{i+1}), \ell^{\mathrm{odd}}(L_{i+1})\big) > U(L_i), \tag{3.1}$$

then the ground state of A_L is simple and even for every $L \in [L_i, L_{i+1}]$.

Proof. For such L, by Lemma 3.1,

$$\lambda_1^{\mathrm{ev}}(L) \leq \lambda_1^{\mathrm{ev}}(L_i) \leq U(L_i) < \min(\ell_2, \ell^{\mathrm{odd}})(L_{i+1}) \leq \min\big(\lambda_2^{\mathrm{ev}}(L), \lambda_1^{\mathrm{odd}}(L)\big);$$

conclude with Corollary 2.3. □

We call the quotient $\min(\ell_2, \ell^{\mathrm{odd}})(L_{i+1})/U(L_i)$ the *step ratio*. Prime-power entries inside a step are harmless, since Lemma 3.1 holds across them.

4. Main theorem

Theorem 4.1. *For every $L \in (0, 2]$ the lowest eigenvalue of the operator A_L associated with Weil's quadratic form (1.1) on $L^2(-L, L)$ is simple, and its eigenfunction is even. Quantitatively:*

1. *on $(0, 0.05]$: $\lambda_2^{\mathrm{ev}} - \lambda_1^{\mathrm{ev}} \geq 0.4172$ and $\lambda_1^{\mathrm{odd}} - \lambda_1^{\mathrm{ev}} \geq 0.6624$;*
2. *on each of the 30 grid steps $[L_i, L_{i+1}] \subset [0.05, 1.19]$ of Table 2: step ratio ≥ 1.288;*
3. *on each of the 8 grid steps $[L_i, L_{i+1}] \subset [1.19, 1.4035]$ of Table 4: step ratio ≥ 1.131;*
4. *on each of the 53 grid steps $[L_i, L_{i+1}] \subset [1.4035, 2]$ of Table 6: step ratio ≥ 2.098.*

Remark 4.2 (Status of release 2.0.0). Items (1) and (2) are the certificate of record of [5], re-checked by the gluing script (§10). Item (3) rests on the 16 lower-bound certificates and 8 upper bounds of Table 4 (release 1.1). Item (4) rests on the 106 lower-bound certificates and 53 upper bounds of Table 6. **One step of item (4) has two proofs.** The pre-registered grid of Piece IV failed as registered at one window, and its registered partial outcome covers $[1.4035, 1.91912841796875]$ (§7.6). The missing step $[1.91912841796875, 1.92364501953125]$ is proved (a) with rule E, a parameter rule registered after that failure, and (b) without rule E, by Lemma 3.1 from the λ_2^{ev} certificate of the registered grid at the next window 1.9326171875, which uses no computation made after the failure (Remark 7.5). Either proof gives item (4) on all of $[1.4035, 2]$. As a pre-registered experiment, the grid nevertheless failed. Piece III continues past 1.4035: its four further steps up to 1.4885 also pass (Table 4, below the rule; smallest ratio 1.129), so $[1.4035, 1.4885]$ is covered by two independent routes; this is not needed for the theorem. At the single window $L = 1.19$ the Piece III machinery also confirms simple-even with a second, independent code path (§6.2). Piece III beyond $L = 1.19$ and Piece IV have been reviewed only by AI agents (§10.3, §11).

The numbers in (1)–(4) are outward-rounded lower ends. The proof is computer-assisted and uses Arb ball arithmetic [20, 34] and exact integer arithmetic; it is given in §5 and, for item (4), in §7.

5. Proof

5.1. **Piece I: small windows** $L \in (0, 0.05]$**.** Here $L < \frac{1}{2}\log 2$, so no prime powers occur. Put $f(x) = L^{-1/2} f_1(x/L)$ with $\|f_1\|_{L^2(-1,1)} = 1$ and $F_1 = \hat{f_1}$. The substitution $t = \tau/L$ gives

$$Q_L(f) = \log\tfrac{1}{L} + R_L(f_1), \qquad R_L(f_1) = P_L(f_1) + \frac{1}{\pi}\int_0^\infty m_L(\tau)|F_1(\tau)|^2\,d\tau,$$

with $m_L(\tau) = \operatorname{Re}\psi(\frac{1}{4} + \frac{i\tau}{2L}) - \log\pi + \log L$, $P_L(f_1) = 2L\big(\int_{-1}^1 f_1 e^{-Ls/2} ds\big)^2$ for even f_1 and $P_L(f_1) = -2L\big(\int_{-1}^1 f_1 \sinh(Ls/2) ds\big)^2$ for odd f_1. The shift $\log(1/L)$ is common to all f_1, so the gaps of Q_L equal those of R_L, sector by sector, and R_L acts on the fixed space $L^2(-1,1)$. With $u = \tau/(2L)$,

$$\partial_L m_L = \tfrac{1}{L} h(u), \qquad h(u) = 1 - \operatorname{Re}\big(iu\,\psi'(\tfrac{1}{4} + iu)\big).$$

Lemma 5.1. *For $u > 0$, $|h(u)| \le 0.8125/u^2$.*

Proof. From $\psi'(w) = \sum_{n\ge 0}(n+w)^{-2}$ one gets $h(u) = 1 - \sum_{n\ge 0}\varphi(n+\frac{1}{4})$ with $\varphi(a) = 2au^2/(a^2+u^2)^2$ and $\int_0^\infty \varphi = 1$. The sum is the midpoint rule for $\int_{-1/4}^\infty \varphi$ with unit cells. The midpoint Peano kernel is bounded by $1/8$ and $\int_{\mathbb{R}} |\varphi''| = 6/u^2$, so the quadrature error is at most $\frac{3}{4}u^{-2}$; and $|\int_{-1/4}^0 \varphi| \le \int_{-1/4}^0 2|a|u^{-2}da = \frac{1}{16}u^{-2}$. □

We bound $\int_0^\infty |h|$ by an upper Riemann sum on $[0, 40]$ with cells of width $1/200$, enclosing $\sup|h|$ on each cell by evaluating h on a ball covering the cell, plus $0.8125/40$ from Lemma 5.1. At 768 bits this gives $\int |h| \le 0.593979$. Since $|F_1|^2 \le (\int |f_1|)^2 \le 2$, the multiplier part of $\frac{d}{dL}R_L(f_1)$ is at most $\frac{4}{\pi}\int|h|$. For the even pole term, with $J = \int f_1 e^{-Ls/2}$, $\frac{d}{dL}(2LJ^2) = 2J^2 + 4LJJ'$ and Cauchy–Schwarz gives $J^2 \le 2\sinh L/L$, $|J'| \le \frac{1}{2}(\frac{2}{3}e^L)^{1/2}$; for the odd pole term, with $K = \int f_1 \sinh(Ls/2)$, $K^2 \le 2\sinh^2(L/2)$ and $|K'| \le \frac{1}{2}\cosh(L/2)\sqrt{2/3}$. All bounds increase in L; evaluated at $L_0 = 0.05$ they give, uniformly for $L \le L_0$,

$$C_{\mathrm{ev}} = \tfrac{4}{\pi}\textstyle\int|h| + 2J^2 + 4L_0|J||J'| \le 4.876362, \qquad C_{\mathrm{odd}} \le 0.761666.$$

The forms R_L share the core $C_c^\infty(-1,1)$ and $|R_L(f_1) - R_{L'}(f_1)| \le C_\sigma|L - L'|\|f_1\|^2$ in sector σ; by min-max every sector eigenvalue of R_L is C_σ-Lipschitz in L. With the certified values $\ell_1, U, \ell_2, \ell^{\mathrm{odd}}$ at $L_0 = 0.05$ (Table 2), for $L \le L_0$,

$$\lambda_2^{\mathrm{ev}}(L) - \lambda_1^{\mathrm{ev}}(L) \ge \ell_2 - U - 2C_{\mathrm{ev}}L_0 \ge 0.4172,$$
$$\lambda_1^{\mathrm{odd}}(L) - \lambda_1^{\mathrm{ev}}(L) \ge \ell^{\mathrm{odd}} - U - (C_{\mathrm{ev}} + C_{\mathrm{odd}})L_0 \ge 0.6624.$$

Both are positive, which proves item (1) of Theorem 4.1. □

Remark 5.2. At 53 bits the same procedure gives the smaller enclosures $\int|h| \le 0.592773$, $C_{\mathrm{ev}} \le 4.87484$, $C_{\mathrm{odd}} \le 0.760130$; at 128 bits $\int|h| \le 0.593583$. All are upper bounds under Arb's containment guarantee and we use the largest. Why the higher-precision upper end is larger is not explained (§11).

5.2. **Per-window certificates on** $[0.05, 1.19]$**.** Fix L and a parity. We use the orthonormal Legendre basis $e_n(x) = c_n P_n(x/L)$, $c_n = \sqrt{(2n+1)/(2L)}$, with $n = 2m$ (even) or $n = 2m+1$ (odd). Up to a unit factor common to the sector, $\hat{e}_n(t) = c_n\, 2L\,(-1)^m j_n(Lt)$ with j_n the spherical Bessel function, and the pole vector is $p_m = \sqrt{2}\, c_n\, 2L\, i_n(L/2)$, so that [pole] $= \pm(p\cdot a)^2$ for $f = \sum a_m e_{n(m)}$.

Lemma 5.3 (Binet bound)**.** *For* $\operatorname{Re} w > 0$, $\left|\psi(w) - \log w + \frac{1}{2w}\right| \le \dfrac{1}{12|w|\operatorname{Re} w}$.

Proof. Binet's formula reads $\psi(w) = \log w - \frac{1}{2w} - \int_0^\infty b(x) e^{-wx} dx$ with $b(x) = \frac{1}{e^x - 1} - \frac{1}{x} + \frac{1}{2}$, $b(0) = 0$. With $y = x/2$, $b'(x) = \frac{1}{4}(y^{-2} - \sinh^{-2} y) \ge 0$. For $y < \sqrt{3}$ the inequality $b' \le \frac{1}{12}$ reduces to $\sinh^2 y \le 3y^2/(3-y^2)$, which holds coefficientwise ($2^{2k-1}/(2k)! \le 3^{1-k}$, with equality at $k = 1, 2$ and term ratio $4/((2k+1)(2k+2)) \le \frac{1}{3}$); for $y \ge \sqrt{3}$ it is trivial. Integrating by parts, $\int b e^{-wx} = w^{-1} \int b' e^{-wx}$, of modulus at most $1/(12|w|\operatorname{Re} w)$. □

Lemma 5.4 (Envelope; improved form of [Zhu, Lemma 3.1])**.** *For* $w = \frac{1}{4} + \frac{it}{2}$ *and* $t \ge 3/2$, $\operatorname{Re}\psi(w) \ge \log\frac{t}{2} - \frac{2}{3t} - \frac{1}{2t^2} \ge \log\frac{t}{2} - \frac{1}{t}$.

Zhu states the envelope for $t \ge 15/4$; the threshold $3/2$ is from the proof below.

Proof. Lemma 5.3 with $|w| \ge t/2$ and $\operatorname{Re}(1/(2w)) \le 1/(2t^2)$; the second inequality is equivalent to $t \ge 3/2$. □

Let $A_L^{\sup} = \sum_{\log n < 2L} 2\Lambda(n)/\sqrt{n}$. The prime comb is $\ge -A_L^{\sup}$ and the envelope increases in t, so for $t \ge T^\sharp$ (Lemma 5.4 needs $T^\sharp \ge 3/2$; every cutoff used is at least 200)

$$\Psi_L(t) \ge \beta := \log\frac{T^\sharp}{2\pi} - \frac{1}{T^\sharp} - A_L^{\sup}. \tag{5.1}$$

Lemma 5.5 (Frequency reduction; [Zhu, Thm. 1.1])**.** *Assume* $\Psi_L \ge \beta > 0$ *on* $[T^\sharp, \infty)$ *and let* $\mathcal{R}(f) = [\text{pole}] + \frac{1}{\pi}\int_0^{T^\sharp} (\Psi_L - \beta)|F|^2 dt + \beta\|f\|^2$. *Then* $Q_L \ge \mathcal{R}$ *as forms, hence* $\lambda_k^\sigma(Q_L) \ge \lambda_k^\sigma(\mathcal{R})$ *for all* k *and* σ.

Proof. Write $\Psi_L = (\Psi_L - \beta) + \beta$, use $\frac{1}{\pi}\int_0^\infty |F|^2 = \|f\|^2$, drop the nonnegative integral over $[T^\sharp, \infty)$ and apply min-max. □

Lemma 5.6 (Monotonicity of $\operatorname{Re}\psi$)**.** *For* $s > 0$, $\operatorname{Re}\psi(s+iy) = \psi(s) + \sum_{n\ge 0} \frac{y^2}{(n+s)((n+s)^2+y^2)}$ *is increasing in* $|y|$. *Consequently* $|\Psi_L - \beta| \le M_\psi := \max(|\operatorname{Re}\psi(\frac{1}{4})|, |\operatorname{Re}\psi(\frac{1}{4} + \frac{iT^\sharp}{2})|) + \log\pi + A_L^{\sup} + \beta$ *on* $[0, T^\sharp]$.

Proof. Take real parts in $\psi(w) = -\gamma + \sum_n \left(\frac{1}{n+1} - \frac{1}{n+w}\right)$. □

Lemma 5.7 (Bessel bounds)**.** $|j_n(x)| \le \dfrac{x^n}{(2n+1)!!}$ *and* $0 \le i_n(x) \le \dfrac{x^n}{(2n+1)!!} e^{x^2/2}$ *for* $x \ge 0$.

Proof. Use Poisson's integral and the beta integral,

$$j_n(x) = \frac{x^n}{2^{n+1} n!} \int_{-1}^{1} \cos(xs)(1-s^2)^n\, ds, \qquad \int_{-1}^{1} (1-s^2)^n\, ds = \frac{2^{2n+1}(n!)^2}{(2n+1)!};$$

for i_n replace cos by cosh $\le e^{x^2/2}$. □

Lemma 5.8 (Gauss–Legendre and Bernstein; [35, Thm. 19.3])**.** *If* g *is analytic in the Bernstein ellipse* E_ρ *with* $|g| \le M$ *there, then the* n*-point Gauss–Legendre rule on* $[-1, 1]$ *has error at most* $\frac{64}{15} M \rho^{-2n+2}/(\rho^2 - 1)$. *A polynomial of degree* d *bounded by* M *on* $[-1, 1]$ *is bounded by* $M\rho^d$ *on* E_ρ.

The cited theorem is stated with ρ^{-2n} for the $(n+1)$-point rule; the form above is the same statement for n points.

Remark 5.9 (Correction of a Gauss–Legendre constant). The original runs of the window certificates applied ρ^{-2n} to the n-point rule, so quadrature balls were too small by ρ^2 (about 4.33 for the head panels, 9 for the Rayleigh outer rule). The error was found in internal review and fixed in the code. The original certificate was first corrected by a Weyl shift of every lower bound by $\delta = N(\rho^2-1)q_{\mathrm{old}}$ ($\delta \le 6.4\cdot 10^{-35}$ for $L \le 0.8$, $\le 1.6\cdot 10^{-33}$ for $0.84 \le L \le 0.97$, $\le 3.2\cdot 10^{-32}$ for $0.98 \le L \le 1$) and fresh upper bounds; that correction assumed unchanged window parameters between code revisions. It has since been superseded by a clean end-to-end rerun with the corrected code, which is the certificate of record used here and does not depend on the correction.

Head matrix. We compute $C_{jk} = \frac{1}{\pi}\int_0^{T^\sharp}(\Psi_L - \beta)\hat e_j \hat e_k\,dt$ by 64-point Gauss–Legendre on each unit panel in Arb. The ellipse has half-height $B = 2/5$, $\rho = 2B + \sqrt{4B^2+1}$. For complex t the continuation of $\operatorname{Re}\psi(\frac14 + \frac{it}{2})$ is $\frac12(\psi(w_+)+\psi(w_-))$, $w_\pm = \frac14 \pm \frac{it}{2}$; on the strip $\operatorname{Re} w_\pm \ge \frac14 - \frac{B}{2} = 0.05$, and Lemma 5.3 bounds $\psi(w_\pm)$. On the strip $|\hat e_j| \le \sqrt{2L}e^{LB}$ and the comb is bounded by $\sum 2\Lambda(n)n^{-1/2}\cosh(B\log n)$. The per-entry error is $T^\sharp\cdot\frac{32}{15}M\rho^{-126}/(\rho^2-1)$ and is added as a ball. The head is $A = C \pm pp^T + \beta I + (\text{quadrature ball})$, with $+$ in the even and $-$ in the odd sector.

Tails. For tail orders $n \ge n_0$ put $X = LT^\sharp$; then $|\hat e_n| \le \sigma_n := c_n 2LX^n/(2n+1)!!$ on $[0,T^\sharp]$ and $\sigma_{n+2}/\sigma_n \le r = \sqrt{\frac{2n_0+5}{2n_0+1}}\frac{X^2}{(2n_0+3)(2n_0+5)} < 1$ (asserted), so $\sum\sigma^2 \le \sigma_{n_0}^2/(1-r^2)$. The tail of the pole vector satisfies $\|p_{\mathrm{tail}}\| \le p_{\mathrm{tail}} := \sqrt{2}c_{n_0}2Ly^{n_0}/(2n_0+1)!!\cdot e^{y^2/2}/\sqrt{1-y^2}$, $y = L/2$. Put

$$e_D = \frac{M_\psi}{\pi}T^\sharp\sum\sigma^2 + [\text{odd}]\,p_{\mathrm{tail}}^2, \qquad e_B = \frac{M_\psi}{\pi}T^\sharp\sqrt{2L}\sqrt{N}\Big(\sum\sigma^2\Big)^{1/2} + 2\|p_{\mathrm{head}}\|p_{\mathrm{tail}}.$$

Then the tail block satisfies $D \ge (\beta - e_D)I$ (in the even sector the tail pole term $+pp^T \ge 0$ is dropped) and the coupling block satisfies $\|X\| \le e_B$ by Hilbert–Schmidt bounds.

Lemma 5.10 (Block bound; [Zhu, Thm. 1.1, §6.2])**.** *Let* $\mathcal{R} = \begin{pmatrix} A & X \\ X^T & D\end{pmatrix}$ *with* $D \ge (\beta - e_D)I$ *and* $\|X\| \le e_B$*. Then* $\lambda_k(\mathcal{R}) \ge \min(\lambda_k(A), \beta - e_D) - e_B$ *for* $1 \le k \le N$*, where* N *is the size of the head* A*.*

Proof. $\left\|\begin{pmatrix} 0 & X \\ X^T & 0\end{pmatrix}\right\| = \|X\|$, so $\mathcal{R} \ge \operatorname{diag}(A,D) - e_B$. The k-th min-max value of $\operatorname{diag}(A,D)$ is at least $\min(\lambda_k(A), \lambda_1(D))$; conclude with Weyl's inequality. □

Verified Cholesky. To show $A - \mu I \succeq 0$ for a ball matrix A, set $d = \mu/100$, compute an approximate Cholesky factor R of $\operatorname{mid}(A) - (\mu+d)I$ in midpoint arithmetic, compute $E = A - (\mu+d)I - RR^T$ rigorously in ball arithmetic, and check $\|E\|_F < d$; then $A - \mu I = RR^T + E + dI \succeq 0$ for every matrix in the ball (cf. [28]). Eigenvalue estimates from inverse iteration on midpoints are used only to choose μ and the deflation vector and carry no rigour. With Lemmas 5.5 and 5.10 this gives $\lambda_1^{\mathrm{ev}} \ge \min(\mu_1, \beta - e_D) - e_B$, $\lambda_1^{\mathrm{odd}} \ge \min(\mu^{\mathrm{odd}}, \beta - e_D^{\mathrm{odd}}) - e_B^{\mathrm{odd}}$, and $\lambda_2^{\mathrm{ev}} \ge \min(\mu_2, \beta - e_D) - e_B$ where μ_2 is certified for $A + vv^T$ (Lemma 5.14 below).

Parameters. The cutoff is $T^\sharp = \max(200, 5\lceil 80/(5L)\rceil, 5\lceil 2\pi e^{A_L^{\sup}+1/2}/5\rceil)$ and the head has $N = \lceil eLT^\sharp/4\rceil + 42$ modes per sector. For $L > 1$ ("sharp cutoff") we use instead $T^\sharp = 2450$ ($L = 1.03$), 4000 ($L = 1.06, 1.09$), 5550 ($L = 1.12$–1.19), the exact dyadic floor $\beta = 3/16$, 112 Gauss nodes per unit panel and $N = \lceil eLT^\sharp/4\rceil + 80$. Lemma 5.4 is applied above the

crude cutoff T_2 (asserting $\log(T_2/2\pi) - 1/T_2 - A_L^{\sup} \geq 3/16$), and $\Psi_L > \beta$ on $[T^\sharp, T_2]$ is certified by Arb enclosures of Ψ_L on cells of width 0.01 with bisection (minimum width 10^{-9}). Together $\Psi_L \geq \beta$ on $[T^\sharp, \infty)$, which is all Lemma 5.5 needs. This device is a refinement inside Zhu's reduction; we make no novelty claim for it. Validation at $L = 1$: the sharp path gives $4.978 \cdot 10^{-30} \leq \lambda_1^{\mathrm{ev}} \leq 9.406 \cdot 10^{-30}$, $\lambda_2^{\mathrm{ev}} \geq 1.808 \cdot 10^{-23}$, $\lambda_1^{\mathrm{odd}} \geq 1.255 \cdot 10^{-26}$, overlapping the crude-cutoff values of Table 2.

Spherical Bessel values for $L \geq 0.98$. Orders $n \leq n_1 = \lfloor x \rfloor$ use upward recurrence in ball arithmetic from Arb seeds every 400 orders; orders $n > n_1$ use the ratios $r_k = j_k/j_{k-1}$ from the backward continued fraction $r_k = x/(2k + 1 - xr_{k+1})$, started from $[0, 1]$ and anchored at j_{n_1}.

Lemma 5.11 (Ratio enclosure)**.** *For $x > 0$ and $k \geq \lfloor x \rfloor + 1$, every true ratio r_k lies in $(0, 1]$; hence the interval iteration started from $[0, 1]$ encloses it.*

Proof. If $2k+1 \geq 2x$, the map $\tau \mapsto x/(2k+1-x\tau)$ sends $[0, 1]$ into $[x/(2k+1), x/(2k+1-x)] \subset [0, 1]$. By downward induction from large K (where $j_K(x) > 0$ and $r_K \in [0, 1]$): if $2k + 1 \geq 2x$ and $r_{k+1} \in [0, 1]$, then $j_{k-1}/j_k = (2k + 1)/x - r_{k+1} \geq 1$, so $r_k \in (0, 1]$ and $j_{k-1} > 0$. The condition holds for all $k \geq \lfloor x \rfloor + 1$. □

The enclosures need not be sharp at the top orders unless the start index is well above $n_{\max}$ [17]; sharpness only affects the size of the radii.

Upper bound U. The trial function is $f = \sum_{m<m^*} v_m e_{2m}$, a polynomial on $[-L, L]$, with v an approximate ground vector (exact dyadic entries). With the rescaled function $f_L(y) := f(Ly)$ on $[-1, 1]$ and $s = x/L \in [0, 2]$, $g(Ls) = L \int_{-1}^{1-s} f_L(y) f_L(y + s)\, dy$ is computed exactly by $(D+1)$-point Gauss–Legendre ($D = \deg f$); the integral in (2.2) by 450-point Gauss–Legendre on $[0, 2]$ with error bounded by Lemma 5.8 ($\rho = 3$, exponent ρ^{-898}, ellipse maximum enclosed by 2048 ball arcs with $|1 - e^{-2Lz}| > 0$ asserted on each arc; the zeros $i\pi k/L$, $k \neq 0$, lie outside the ellipse); the other terms of (2.2) exactly. Then $U \geq Q_L(f)/\|f\|^2 \geq \lambda_1^{\mathrm{ev}}(L)$. A polynomial on $[-L, L]$ has $|\hat{f}|^2 = O(t^{-2})$ and lies in the form domain $\{f : \int \log(2 + t)|\hat{f}|^2 < \infty\}$, and in the closure of $C_c^\infty(-L, L)$ there, since the weight is weaker than any H^s, $s < \frac{1}{2}$.

5.3. **Piece II: the grid on $[0.05, 1.19]$.** The grid is

0.05, 0.1, 0.2, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.64, 0.68, 0.72, 0.76, 0.8, 0.84,
0.87, 0.9, 0.93, 0.96, 0.97, 0.98, 0.99, 1.0, 1.03, 1.06, 1.09, 1.12, 1.15, 1.18, 1.19.

For each of the 30 steps the criterion (3.1) holds (Table 2); the smallest ratios are 1.288 on $[0.05, 0.1]$, 1.320 on $[0.1, 0.2]$ and, past $L = 1$, 2.148 on $[1.15, 1.18]$. By Corollary 3.2 this proves item (2) of Theorem 4.1. □

Remark 5.12 (Comparisons and dyadic windows). (a) The gluing script compares balls with Python's built-in `min`, which is not a rigorous minimum when balls overlap; here the two lower bounds are disjoint at every window (relative separation ≥ 1.021, radii $< 5 \cdot 10^{-20}$), and the step ratios in Table 2 were recomputed with explicit endpoint comparisons. (b) Each window is certified at the dyadic number $\tilde{L}_i \geq L_i$ with $\tilde{L}_i < L_i + 2^{-p}$, $p \geq 768$, for all four bounds of the window; the covering is then $(0, \tilde{L}_1] \cup \bigcup_i [\tilde{L}_i, \tilde{L}_{i+1}] \supset (0, 1.19]$.

5.4. **Piece III: the grid on $(1.19, 1.4035]$.** Beyond $L \approx 1.2$ the route of §5.2 is impractical: the cutoff $T^\sharp \gtrsim 2\pi e^{A_L^{\sup}}$ forced by the pointwise bound (5.1) is about $2 \cdot 10^5$ at $L = 1.3$ and $5 \cdot 10^6$ at $L = 1.5$, and Zhu [Lemma 3.2] shows that $A_L^{\sup}$ is optimal for any pointwise bound. Elston [16] replaces the pointwise bound by an operator bound for the prime comb on a slightly

widened window. We state his reduction as a cited proposition; we use his code unchanged except for the deflation option described below.

Notation. Let $a = L$ be the window, $\delta > 0$, $b = a + \delta$, $\beta_K > 0$, and $T^\sharp > \beta_K/\delta$ with $\Omega := T^\sharp - \beta_K/\delta$ and $\Omega\delta \geq 1$. Let $W(u) = I_0(\beta_K\sqrt{1-u^2/\delta^2})/I_0(\beta_K)$ on $[-\delta,\delta]$ (Kaiser window, 0 outside), $\kappa(u) = \frac{\sin \Omega u}{\pi u} W(u)$ and $m = 1 - \hat{\kappa}$. Let $H(t) = \operatorname{Re}\psi(\frac{1}{4} + \frac{it}{2}) - \log\pi$, $P(t) = \sum_{\log n<2a} \frac{2\Lambda(n)}{\sqrt{n}} \cos(t\log n)$, so $\Psi_a = H - P$, and $A = \sum_{\log n<2a} 2\Lambda(n)/\sqrt{n}$. Let $C_b = \sum_{\log n<2a} \Lambda(n) n^{-1/2}(\tau_{\log n} + \tau_{-\log n})$, compressed to $L^2(-b,b)$, where τ_s is translation by s.

Proposition 5.13 (Elston [16])**.** *Assume*

(H1) $\bar{\mu} \geq \sup\{\langle h, C_b h\rangle/\|h\|^2 : 0 \neq h \in L^2(-b,b)\}$;
(H2) $\varepsilon \geq \sup_{t\geq T^\sharp} |\hat{\kappa}(t)|$ *(in* [16]*,* $\varepsilon = 2/(\pi\beta_K I_0(\beta_K)))$*;*
(H3) $0 < \beta' \leq \log\frac{T^\sharp}{2\pi} - \frac{1}{T^\sharp} - \bar{\mu} - (2\varepsilon + \varepsilon^2)(A + \bar{\mu})$.

Put $\Phi = \Psi_a + m^2(P - \bar{\mu})$ *and*

$$R'(f) = [\text{pole}] + \frac{1}{\pi}\int_0^{T^\sharp} (\Phi - \beta')|F|^2\,dt + \beta'\|f\|^2.$$

Then:

(a) $Q_a \geq R'$ *as forms on each parity sector.*
(b) *Let* A_N *be the matrix of* R' *on the first* N *Legendre modes of the sector, and let* $\varepsilon_D, \varepsilon_B$ *be the tail constants of Zhu's reduction computed with* $G \geq \sup_{[0,T^\sharp]} |\Phi - \beta'|$ *(Bessel tail bounds as in Lemma 5.7;* ε_D *by a Gershgorin bound on the tail rows,* ε_B *by a Schur test). Then the tail block of* R' *is* $\succeq (\beta' - \varepsilon_D)I$ *and the coupling block has norm* $\leq \varepsilon_B$.
(c) *The certifier encloses every entry of* A_N *in a ball (Gauss–Legendre nodes and weights,* Ψ *via the complex digamma function,* m *by rigorous integration of the Kaiser main lobe and a mean-value bound for the side lobes,* j_n *by recurrence and ratio enclosure), adds a spectral quadrature debit* c_{err} *from a Bernstein-ellipse bound on panels of width* $1/4$*, and certifies a lower bound* $\lambda_1(A_N) \geq \mu' - \rho_{\mathrm{res}} - N r_{\max} - c_{\mathrm{err}}$*, where* μ' *is a dyadic shift,* ρ_{res} *the* ∞*-norm of an exactly computed integer Cholesky residual divided by the scale* 2^{2P} *(plus* $N2^{-2P}$ *for the floors), and* $r_{\max}$ *the largest entry radius.*

Proof of (a), following [16]. Put $h = f - f * \kappa$; then $\operatorname{supp} h \subset [-b,b]$ and $\hat{h} = mF$. By (H1), $\frac{1}{\pi}\int_0^\infty Pm^2|F|^2 = \langle h, C_b h\rangle \leq \bar{\mu}\|h\|^2 = \frac{1}{\pi}\int_0^\infty \bar{\mu} m^2|F|^2$, so $\frac{1}{\pi}\int m^2(\bar{\mu} - P)|F|^2 \geq 0$ and $Q_a(f) \geq [\text{pole}] + \frac{1}{\pi}\int_0^\infty \Phi|F|^2$. For $t \geq T^\sharp$, $\Phi = H - \bar{\mu} - (1-m^2)(P - \bar{\mu})$ with $|1-m^2| \leq 2\varepsilon + \varepsilon^2$ by (H2) and $|P - \bar{\mu}| \leq A + \bar{\mu}$, so Lemma 5.4 and (H3) give $\Phi \geq \beta'$; dropping $\frac{1}{\pi}\int_{T^\sharp}^\infty (\Phi - \beta')|F|^2 \geq 0$ gives $Q_a \geq R'$. □

Elston states (b)–(c) for the lowest eigenvalue; in [16], $\bar{\mu}$ is certified by a weighted Schur test on 4000–8000 cells with a Collatz–Wielandt weight, and the finite certificate is an exact integer Gram residual in the spirit of [28]. An independent audit of the code at $L = 1.19$ (documented in [16]) found and fixed two implementation issues; the code used here is the post-audit version (upstream commit `3cc4cb7b`).

Lemma 5.14 (Rank-one deflation)**.** *Let* A *be a real symmetric* $N \times N$ *matrix and* $w \in \mathbb{R}^N$*. Then* $\lambda_2(A) \geq \lambda_1(A + ww^T)$*. In particular, if* $A + ww^T - \mu I \succeq 0$ *then* A *has at most one eigenvalue below* μ.

Proof. By Courant–Fischer, $\lambda_2(A) = \max_{y\neq 0} \min\{x^T A x : \|x\| = 1,\, x \perp y\} \geq \min\{x^T A x : \|x\| = 1,\, x \perp w\} = \min\{x^T(A + ww^T)x : \|x\| = 1,\, x \perp w\} \geq \lambda_1(A + ww^T)$. (For $w = 0$ the claim is trivial.) □

Corollary 5.15 (Second even eigenvalue)**.** *Under the hypotheses of Proposition 5.13, in the even sector and for any $w \in \mathbb{R}^N$,*

$$\lambda_2^{\mathrm{ev}}(L) \;\geq\; \lambda_2(R'^{\mathrm{ev}}) \;\geq\; \min\bigl(\lambda_1(A_N + ww^T),\ \beta' - \varepsilon_D\bigr) - \varepsilon_B.$$

Proof. By Lemma 5.14, $\lambda_2(A_N) \geq \lambda_1(A_N + ww^T)$. Lemma 5.10 with $k = 2$ (Weyl's inequality applied to the off-diagonal blocks, with the bounds of Proposition 5.13(b)) gives $\lambda_2(R') \geq \min(\lambda_2(A_N), \beta' - \varepsilon_D) - \varepsilon_B$. Since $Q_L \geq R'$ as forms (Proposition 5.13(a)), min-max gives $\lambda_2^{\mathrm{ev}}(L) \geq \lambda_2(R'^{\mathrm{ev}})$. □

Implementation of the deflation (option `--deflate`; our only change to [16]). After the ball matrix A_N is assembled, a vector v approximating the unit lowest eigenvector of $\mathrm{mid}(A_N)$ is computed by 12 steps of inverse iteration; its entries are rounded to 20 significant decimal digits and converted to Arb balls. The matrix $A_N + vv^T$ is formed in ball arithmetic, so it encloses $A_N + ww^T$ for the exact midpoint vector w of v, and the rounding of v enters the largest entry radius $r_{\max}$ exactly like the Gram radii. The certifier then runs unchanged on $A_N + vv^T$, and Proposition 5.13(c) yields a certified lower bound for $\lambda_1(A_N + ww^T)$. The choice of v affects only sharpness, not rigour.

One form, two codes. The step test (3.1) compares lower bounds from Elston's certifier with upper bounds from our code, so both must refer to the same form with the same norm. Both implement (1.1) in Zhu's normalisation ($\|f\|^2 = \frac{1}{\pi}\int_0^\infty |F|^2$ for real f, pole term $+2F(\frac{i}{2})^2$ in the even and $-2|F(\frac{i}{2})|^2$ in the odd sector). As a consistency check, at $L = 1.19$ Elston's certified even lower bound $6.8131 \cdot 10^{-48}$, our Piece II bracket $[7.8 \cdot 10^{-48}, 1.312 \cdot 10^{-47}]$, the Legendre–Galerkin bound $U(1.19) \leq 8.712 \cdot 10^{-48}$ ($K = 640$, below) and the non-rigorous split-window value $8.676 \cdot 10^{-48}$ (basis of §9) are mutually compatible; in the odd sector Elston's certified lower bound $3.9196 \cdot 10^{-44}$ lies below our rigorous Galerkin upper bound $4.8461 \cdot 10^{-44}$ ($K = 256$). The proof does not rely on these numerical agreements; it relies only on both codes evaluating (1.1).

Upper bounds $U(L_i)$. At every grid point L_i, including $L_i = 1.19$, $U(L_i)$ is the outward-rounded upper endpoint of a Rayleigh quotient $v^T Q_K v / v^T v$ of the even Legendre–Galerkin matrix Q_K ($K = 640$ even modes, 3000 bits) with an exact dyadic vector v, computed in ball arithmetic with the rigorous quadrature bound of §8 (Proposition 8.1). At $L = 1.19$ this gives $U(1.19) \leq 8.712 \cdot 10^{-48}$, sharper than the $1.312 \cdot 10^{-47}$ of Piece II (Table 2). The sharper value is needed: with $1.312 \cdot 10^{-47}$ the first step ratio would be 0.93. Both are valid upper bounds, and the gluing script uses the smaller one at each window.

Grid. The planned grid to $L = 1.5$ is

$$1.19,\ 1.2205,\ 1.25,\ 1.2785,\ 1.3055,\ 1.3315,\ 1.3565,$$
$$1.3805,\ 1.4035,\ 1.4255,\ 1.447,\ 1.468,\ 1.4885,\ 1.5.$$

It was chosen from non-rigorous Galerkin estimates so that $0.95 \min(\lambda_2^{\mathrm{ev}}, \lambda_1^{\mathrm{odd}})(L_{i+1}) \geq 1.25 \cdot 1.03 \cdot \lambda_1^{\mathrm{ev}}(L_i)$. The windows $1.2205, \ldots, 1.4885$ are complete in both sectors; the runs at 1.5 were not complete when this version was prepared and are not used. Only the windows up to 1.4035 enter the theorem; $[1.4035, 2]$ is Piece IV. The parameters actually used are in Table 3 and in the field `args` of every certificate file. They follow a design rule (`code/gap/params.py`) that plays no role in the proof: $\Omega = T^\sharp - \beta_K/\delta$ is interpolated log-linearly from Elston's runs at $L = 1.19, 1.3, 1.495$ and inflated by 3%; $N \approx 0.8L(\Omega + 64/\delta) + 20$; the number of Gauss points per panel is raised so that the quadrature debit is at most 10^{-3} times the target eigenvalue (computed in the certifier's plan mode); and β_K is the smallest value for which N times the side-lobe entry radius stays below 10^{-6} times the target eigenvalue. The last rule was added

after a first pass with $\beta_K = 64$ at every window gave a negative odd bound at 1.2785; the runs from that pass that gave useful bounds (1.2205, 1.25, and the even sector at 1.2785) are kept. Grid values are passed to the certifier as decimal strings, converted to Arb balls that contain the exact decimal; the Schur window b is rounded up, which is admissible because the supremum in (H1) is monotone in b.

Conclusion of Piece III. Each of the 8 steps from 1.19 to 1.4035 satisfies (3.1) (Table 4); the smallest ratio is 1.131, on $[1.2205, 1.25]$, and the odd sector binds at every window. By Corollary 3.2 this proves item (3) of Theorem 4.1. □

The four further steps $[1.4035, 1.4255]$, $[1.4255, 1.447]$, $[1.447, 1.468]$ and $[1.468, 1.4885]$ also satisfy (3.1) (Table 4, below the rule; smallest ratio 1.129, on $[1.468, 1.4885]$), so Piece III alone gives a simple, even ground state on $[1.19, 1.4885]$. Piece IV covers $[1.4035, 2]$ by a different method, so these steps are not needed for Theorem 4.1; they serve as the independent cross-check of §7.8.

5.5. **Piece IV: the grid on** $[1.4035, 2]$**.** Item (4) of Theorem 4.1 is proved in §7 (Corollary 3.2 applied to the 53 steps of Table 6). Since the four pieces share their endpoints 0.05, 1.19 and 1.4035, they cover $(0, 2]$. □

6. Results

6.1. **Pieces I and II.** Table 2 lists the certificate of record on $[0.05, 1.19]$. Lower bounds are rounded down and upper bounds up, to four significant digits; the step ratio in row L_{i+1} is $\min(\ell_2, \ell^{\mathrm{odd}})(L_{i+1})/U(L_i)$, lower end floored. The odd sector binds at every window from $L = 0.1$ on. Piece I on this certificate: $\lambda_2^{\mathrm{ev}} - \lambda_1^{\mathrm{ev}} \geq 0.4172$ and $\lambda_1^{\mathrm{odd}} - \lambda_1^{\mathrm{ev}} \geq 0.6624$ on $(0, 0.05]$.

At $L = 0.8$ our certified values are consistent with Zhu's [Thm. 6.2]: $8.9 \cdot 10^{-18} \leq \lambda_1^{\mathrm{ev}} \leq 2.523 \cdot 10^{-16}$, $\lambda_2^{\mathrm{ev}} \geq 2.085 \cdot 10^{-12}$, $8.206 \cdot 10^{-15} \leq \lambda_1^{\mathrm{odd}} \leq 2.347 \cdot 10^{-14}$. Zhu's certified upper bound $\lambda_1^{\mathrm{ev}}(0.8) \leq 2.27 \cdot 10^{-17}$ [Table 3] is sharper than our $3.681 \cdot 10^{-17}$.

6.2. **Piece III.** *Test at* $L = 1.19$*.* With Elston's published parameters at $L = 1.19$ ($\delta = 0.3$, $\beta_K = 64$, $T^\sharp = 700$, $N = 680$, 56 Gauss points on panels of width 1/4) the certifier, run by us, gives

$$\lambda_1^{\mathrm{ev}}(1.19) \geq 6.8131164696015 \cdot 10^{-48}, \qquad \lambda_1^{\mathrm{odd}}(1.19) \geq 3.91966861609542239 \cdot 10^{-44},$$

reproducing Elston's published $6.8131164696007 \cdot 10^{-48}$ (even, 12 digits) and his odd bound to 17 digits; the shift μ_s depends on the thread count through the numerical eigenvalue estimate, and both values are valid lower bounds. With `--deflate`,

$$\lambda_2^{\mathrm{ev}}(1.19) \geq 1.2336395547 \cdot 10^{-40}$$

(non-rigorous Galerkin value $1.23364079 \cdot 10^{-40}$). With $U(1.19) \leq 1.312 \cdot 10^{-47}$ from Table 2, $\min(\lambda_2^{\mathrm{ev}}, \lambda_1^{\mathrm{odd}})(1.19)/U(1.19) \geq 2987$: simple-even at $L = 1.19$ is confirmed by a second, independent code path (Elston's reduction instead of Zhu's pointwise cutoff, a different Gram assembly and a different Cholesky certificate).

7. Piece IV: the Schur-certificate route on $[1.4035, 2]$

Remark 7.1 (Disclosure: registered failure and post-hoc repair). The grid of this piece was pre-registered with a hash stamp before any grid computation. **As registered, it failed** at one window ($L = 1.92364501953125$, no certified λ_2^{ev} bound at the planned cut N). The registered partial outcome covers $[1.4035, 1.91912841796875]$. The gap was closed by *rule E* ($N \to N + 1024$), which was written and registered *after* that failure and after a diagnostic run had passed at the new N; the scored run under rule E passed. The one step that the

TABLE 2. Certificate of record on $[0.05, 1.19]$. $T^\sharp$ and N are the cutoff and the head size per sector. At $L = 1$ the crude cutoff is used (the sharp-path validation is in §5.2).

L	$T^\sharp$	N	$\lambda_1^{\mathrm{ev}} \geq$	$\lambda_1^{\mathrm{ev}} \leq U$	$\lambda_2^{\mathrm{ev}} \geq$	$\lambda_1^{\mathrm{odd}} \geq$	step ratio $\geq$
0.05	1600	97	0.9351	0.9827	1.887	1.926	(Piece I)
0.1	800	97	0.4348	0.4572	1.387	1.266	1.288
0.2	400	97	0.09014	0.09499	1.046	0.6037	1.320
0.3	270	98	7.143e-3	7.587e-3	0.7437	0.2110	2.222
0.35	230	97	1.110e-3	1.204e-3	0.6153	0.06086	8.022
0.4	200	97	1.662e-4	1.841e-4	0.2866	0.01361	11.31
0.45	200	104	1.465e-5	1.647e-5	0.08888	2.205e-3	11.98
0.5	200	110	8.482e-7	9.506e-7	0.01650	1.752e-4	10.63
0.55	200	117	4.424e-8	6.066e-8	1.234e-3	1.213e-5	12.76
0.6	200	124	1.321e-9	1.813e-9	8.661e-5	4.921e-7	8.114
0.64	200	129	7.330e-11	1.036e-10	5.914e-6	3.008e-8	16.59
0.68	200	135	1.984e-12	2.832e-12	3.870e-7	1.180e-9	11.39
0.72	200	140	5.422e-14	1.207e-13	1.013e-8	3.167e-11	11.18
0.76	200	146	7.533e-16	1.802e-15	3.574e-10	7.456e-13	6.181
0.8	200	151	9.787e-18	3.681e-17	4.895e-12	1.086e-14	6.031
0.84	830	516	1.134e-19	1.757e-19	8.754e-14	1.322e-16	3.593
0.87	830	533	2.636e-21	4.329e-21	2.568e-15	3.892e-18	22.15
0.9	830	550	3.401e-23	5.545e-23	5.866e-17	5.819e-20	13.44
0.93	830	567	4.303e-25	7.594e-25	8.697e-19	9.536e-22	17.20
0.96	830	584	4.510e-27	7.555e-27	9.996e-21	8.439e-24	11.11
0.97	830	590	8.936e-28	1.613e-27	2.109e-21	1.929e-24	255.3
0.98	3610	2447	1.682e-28	2.165e-28	5.262e-22	4.471e-25	277.2
0.99	3610	2471	2.782e-29	3.552e-29	1.101e-22	7.684e-26	354.9
1.0	3610	2496	4.531e-30	6.601e-30	1.649e-23	1.141e-26	321.5
1.03	2450	1795	2.096e-32	4.110e-32	1.055e-25	6.604e-29	10.00
1.06	4000	2962	7.275e-35	1.249e-34	4.435e-28	2.691e-31	6.548
1.09	4000	3043	1.306e-37	2.162e-37	1.034e-30	4.711e-34	3.774
1.12	5550	4305	1.903e-40	3.001e-40	2.049e-33	8.502e-37	3.933
1.15	5550	4418	1.752e-43	2.799e-43	2.501e-36	8.572e-40	2.857
1.18	5550	4531	9.792e-47	1.636e-46	2.088e-39	6.011e-43	2.148
1.19	5550	4569	7.823e-48	1.312e-47	1.439e-40	4.329e-44	264.7

registered grid left open, $[1.91912841796875, 1.92364501953125]$, also follows without rule E, by monotonicity from a registered record at the next window (Remark 7.5). Details are in §7.6.

7.1. **Origin of the method, credit and clean-room protocol.** The method of this piece is bramble_trace's [8]: summed digamma responses, a full matrix of displacement rank 2 in a trigonometric basis, exact elimination of a finite core through rational functions, a moment and Abel-summation bound for the far block, and a Hilbert–Schmidt bound for the omitted tail. The Green-response identity, the trigonometric bases and the closed-form comparison are lihongmin's [23]. Both are AI-agent preprints on rStation, accepted only by AI reviewer agents and not refereed. They prove positivity of both parity sectors, [8] for every support radius $0 < a \leq 2$ and [23] for $0 < a \leq \frac{1}{2}\log 11$, without eigenvalue lower bounds and without a statement on the ground state (§1.3). Our clean-room code independently re-checks these positivity statements at our windows (§1.3).

Clean-room protocol. (i) A specification and a proof document were written from the manuscript text of [23, 8] only; no supplement code of either paper was opened (the code licence

TABLE 3. Piece III windows and certifier parameters, from the field `args` of each certificate file. Common to all runs: panel width 1/4, Schur test on 4000 cells with 1500 weight iterations. "GL" is the number of Gauss–Legendre points per panel, "bits" the working precision, "time" the wall-clock seconds (field `seconds`) of the λ_2^{ev} (deflated) and λ_1^{odd} runs, with 11–52 worker processes. At 1.2785 the two sectors used different parameters. New prime powers relative to the previous row are listed. The rows below the rule (windows past 1.4035, not used in the proof) are generated from the certificate files by `paper/gen_tables.py`.

L_i	δ	β_K	$T^\sharp$	N	GL	bits	time (s), λ_2^{ev} / λ_1^{odd}	new n
1.19	0.3	64	700	680	56	400	600–720 each	(test; $n \le 9$)
1.2205	0.3	64	850	849	56	400	1009 / 714	11
1.25	0.3	64	990	1010	56	400	1481 / 1034	
1.2785, even	0.3	64	1150	1196	57	400	2418 / —	
1.2785, odd	0.3	74	1130	1133	63	400	— / 1000	
1.3055	0.3	78	1290	1315	65	400	3301 / 2466	13
1.3315	0.2	84	1570	1585	70	400	5673 / 4384	
1.3565	0.2	89	1730	1757	74	600	11116 / 8545	
1.3805	0.2	93	1890	1943	78	600	11639 / 7998	
1.4035	0.2	98	2070	2143	81	600	14363 / 9784	16
1.4255	0.2	102	2240	2356	85	600	18715 / 13217	17
1.447	0.2	107	2440	2587	89	600	24322 / 20970	
1.468	0.2	111	2640	2837	92	600	31456 / 25765	
1.4885	0.2	115	2850	3107	95	600	40211 / 44638	19

of [8] allows execution and inspection for verification only, and [23] reserves all rights). (ii) The implementer worked from these two documents and our own code only, and opened no rStation file. (iii) No rStation code was copied, adapted or vendored; an independent audit searched our code for the manuscripts' code identifiers and characteristic constants and found none. (iv) Every step was re-derived from first principles in the proof document, including steps that [8] spot-checks only. Since this version the proof document and the specification are part of the release as a supplement [7]: `supplement/PROOFS.md` (items P1–P23, byte-identical to the file whose hash is in the run-time stamp `data/piece4/GAP.stamp`), `supplement/SPEC.md` (a copy with one host name redacted) and `supplement/MAP.md`, which maps every formula to the code and to the fields of the run records. They contain no rStation code. The shipped code of this piece is ours, apart from Elston's MIT-licensed integer Cholesky and weighted Schur-test routines [16], which Piece III already uses.

Our additions. (a) All formulas at general L ([8] works at $a = 2$). (b) A spectral shift μ, which turns the positivity certificate into a *value* certificate $\lambda_k^\sigma(L) \ge \mu$. (c) The rank-one deflation of Lemma 5.14, lifted to the operator; it adds a second generator pair and raises the displacement rank of the core Schur complement from 2 to 4. (d) Lemma 7.4, an exact-integer structured LDL^{T} factorisation with an explicit residual bound, in place of the floating-point factorisation and rounding lemma of [8]. (e) A form-core argument in place of the H^s-cutoff density argument, and a proof of the far-block diagonal correction by virtual labels. (f) Independent per-run gates, among them G0i and C-ind (§7.4). (g) Pre-registration, rule E, and the gluing with Pieces I–III.

7.2. **The form in a trigonometric basis.** Fix L and a sector $\varepsilon = +1$ (even) or -1 (odd), and put $\omega = 0$ or $\frac{1}{2}$, $k_q = q + \omega$, $\kappa_q = \pi k_q / L$ and the labels $\lambda_q = k_q^2$ ($q \ge 0$). On $(0, L)$ let

TABLE 4. Piece III certificate. $U(L_i) \geq \lambda_1^{\mathrm{ev}}(L_i)$ is the $K = 640$ Galerkin bound; ℓ^{odd} and ℓ_2 are certified lower bounds for λ_1^{odd} and λ_2^{ev}; the step ratio in row L_{i+1} is $\min(\ell_2, \ell^{\mathrm{odd}})(L_{i+1})/U(L_i)$. Lower bounds are rounded down and upper bounds up, to four significant digits, and ratios are floored. The lower bounds in row 1.19 are the test of §6.2 and enter no step. Below the rule: the further steps of Piece III past 1.4035, generated from the certificate files by `paper/gen_tables.py`; they are also covered by Piece IV and are not used in the proof of Theorem 4.1.

L_i	$U(L_i)$	$\lambda_1^{\mathrm{odd}} \geq$	$\lambda_2^{\mathrm{ev}} \geq$	step ratio $\geq$
1.19	8.712e-48	3.919e-44	1.233e-40	(Piece II: 264.7)
1.2205	2.336e-51	1.224e-47	4.200e-44	1.405
1.25	4.458e-55	2.642e-51	1.162e-47	1.131
1.2785	8.317e-59	5.889e-55	2.648e-51	1.321
1.3055	1.209e-62	9.466e-59	4.806e-55	1.138
1.3315	1.693e-66	1.706e-62	9.901e-59	1.412
1.3565	1.962e-70	2.187e-66	1.435e-62	1.292
1.3805	2.251e-74	3.032e-70	2.247e-66	1.546
1.4035	2.493e-78	3.492e-74	2.991e-70	1.552
1.4255	2.371e-82	3.564e-78	3.366e-74	1.429
1.447	2.107e-86	3.269e-82	3.420e-78	1.379
1.468	1.740e-90	2.834e-86	2.886e-82	1.345
1.4885	1.084e-94	1.964e-90	2.141e-86	1.129

$\tilde{e}_q = (-1)^q\sqrt{2/L}\,\gamma_q\cos(\kappa_q x)$ (even; $\gamma_0 = 2^{-1/2}$, $\gamma_q = 1$ for $q \geq 1$) or $\tilde{e}_q = (-1)^q\sqrt{2/L}\sin(\kappa_q x)$ (odd), and let $\mathcal{U}_\varepsilon v(x) = v(|x|)/\sqrt{2}$ for $x > 0$ and $\varepsilon v(|x|)/\sqrt{2}$ for $x < 0$, a unitary map onto the sector. Put $H(t) = \operatorname{Re}\psi(\frac{1}{4} + \frac{it}{2}) - \log\pi$ and $c_j = 2j + \frac{1}{2}$.

Proposition 7.2 (Green expansion; cf. [23, Lemma 3.1])**.** *For $f = \mathcal{U}_\varepsilon v \in C_c^\infty(-L, L)$ of parity ε and $x_q = \langle v, \tilde{e}_q\rangle$,*

$$Q_L(f) = \sum_{q\geq 0} H(\kappa_q)x_q^2 - \langle C_L f, f\rangle + \sum_{j\geq 0}\langle r_j, v\rangle^2 + \varepsilon\langle p_\varepsilon, v\rangle^2,$$

where $C_L = \sum_{\log n<2L}\Lambda(n)n^{-1/2}(\tau_{\log n} + \tau_{-\log n})$ compressed to $L^2(-L, L)$, τ_s is translation by s, $r_j(x) = e^{-c_jL}(e^{c_jx} + \varepsilon e^{-c_jx})/\sqrt{1 - \varepsilon e^{-2c_jL}}$ and $p_\varepsilon(x) = e^{x/2} + \varepsilon e^{-x/2}$.

Sketch. From $\psi(w) = -\gamma + \sum_n(\frac{1}{n+1} - \frac{1}{n+w})$, $H(t) - H(0) = \sum_j[2/c_j - 2c_j/(c_j^2 + t^2)]$, and $2c/(c^2 + t^2)$ is the Fourier transform of $e^{-c|x|}$. The resolvent $R_c = (-\partial^2 + c^2)^{-1}$ on $(0, L)$ with $v'(L) = 0$ and $v'(0) = 0$ (even) or $v(0) = 0$ (odd) has eigenbasis $\tilde{e}_q$ with eigenvalues $(c^2 + \kappa_q^2)^{-1}$, and its kernel satisfies $2cR_c - E_c = r_c \otimes r_c$ with $E_c(x, y) = e^{-c|x-y|} + \varepsilon e^{-c(x+y)}$ (direct computation with the Green function). Summing over j (Tonelli) gives the archimedean part; the prime and pole terms are those of (1.1). □

The matrix $A = (A_{qr})$ of Q_L in the basis $\mathcal{U}_\varepsilon\tilde{e}_q$ has displacement rank 2: with $D_\lambda = \operatorname{diag}(\lambda_q)$, $D_\lambda A - AD_\lambda = uv^{\mathsf{T}} - vu^{\mathsf{T}}$, where $v_q = \gamma_q$, $u_q = \gamma_q(2L/\pi^2)\Phi_\varepsilon(\kappa_q)$,

$$\Phi_\varepsilon(x) = x\sum_{\log n<2L}\frac{\Lambda(n)}{\sqrt{n}}\sin(x\log n) + G(x) - \varepsilon x^2\sum_{j\geq 0}\frac{e^{-2c_jL}}{c_j^2 + x^2} - \frac{\alpha_\varepsilon}{\frac{1}{4} + x^2},$$

$G(x) = \frac{x}{2}\operatorname{Im}\psi(\frac{1}{4}+\frac{ix}{2})$, $\alpha_+ = \sinh^2\frac{L}{2}$ and $\alpha_- = -\cosh^2\frac{L}{2}$ [7, P4–P5]. With $w_n = \Lambda(n)n^{-1/2}$, $S(x) = \sum_{\log n<2L} w_n \sin(x\log n)$ and $C(x) = \sum_{\log n<2L} w_n\big(1-\frac{\log n}{2L}\big)\cos(x\log n)$, the diagonal is [7, P6]

$$A_{qq} = H(\kappa_q) - 2C(\kappa_q) + [\kappa_q > 0]\,\frac{S(\kappa_q)}{L\kappa_q} + \frac{2}{L}\gamma_q^2 R_\varepsilon(\kappa_q),$$

$$R_\varepsilon(x) = \frac{\operatorname{Im}\psi(\frac{1}{4}+\frac{ix}{2})}{4x} + \frac{\operatorname{Re}\psi'(\frac{1}{4}+\frac{ix}{2})}{8} - \varepsilon\sum_{j\ge0}\frac{c_j^2 e^{-2c_jL}}{(c_j^2+x^2)^2} + \frac{\alpha_\varepsilon}{(\frac{1}{4}+x^2)^2},$$

and at even $q = 0$ it reads $A_{00} = H(0) - 2C(0) + R_+(0)/L$ with $R_+(0) = \psi'(\frac{1}{4})/4 - \sum_j e^{-2c_jL}/c_j^2 + 16\alpha_+$. All entries are evaluated in Arb from these closed forms, with an explicit ball for the truncation of the sums over j; no quadrature enters. The closed forms are checked against the position-side formula (2.2) at every window (gates G0, G0-mini and G0i below).

7.3. **The value certificate.** Split the indices into a core $[0,K)$, an intermediate block $I_P = [K,N)$, a finite far block $[N,M)$ and the tail $q \ge M$. Fix an exact dyadic $\mu \ge 0$; in the even sector also a deflation weight $\vartheta \ge 0$ and a dyadic $\phi \in \mathbb{Q}^K$ (a rounded approximate lowest eigenvector of the core), and put $\Phi = \mathcal{U}_\varepsilon \sum_{q<K}\phi_q\tilde{e}_q$; in the odd sector $\vartheta = 0$.

Theorem 7.3 (Value certificate; general-L, shifted and deflated form of [8, (9.1)–(9.2)])**.** *Assume:*

- (V1) $s \ge \sup\{\langle C_b h,h\rangle/\|h\|^2 : 0 \ne h \in L^2(-b,b)\}$ *for a dyadic* $b \ge L$, *with* C_b *as in Proposition 5.13;*
- (V2) $\mathsf{M} := A_{\mathrm{core}} - \mu I + \vartheta\phi\phi^{\mathsf{T}} \succ 0$;
- (V3) $t'_N > 0$, *where the dyadic numbers* t'_q *satisfy* $t'_N \le t'_q \le H(\kappa_q) - \mu - s - \tau_N$ *for* $q \ge N$, *and* τ_N *bounds the far block of the core Schur complement (plus, in the odd sector, the far part of the pole term);*
- (V4) *Lemma 7.4 applied to* $S_{\mathrm{input}} - \delta_0 I$ *completes with positive pivots;*
- (V5) $\delta_0 - \varepsilon_E - \varepsilon_{\mathrm{asm}} - r_M > 0$.

Here S_{input} *is a dyadic rounding of the Schur complement* S_M *on* I_P *that remains after eliminating the core exactly and the finite far block against* $\operatorname{diag}(t'_q)$, $\varepsilon_{\mathrm{asm}} \ge \|S_M - S_{\mathrm{input}}\|$, ε_E *is the residual bound of Lemma 7.4, and* r_M *bounds the omitted tail* $q \ge M$. *Then the closure* $\bar{Q}_L$ *of* Q_L *satisfies the* certificate inequality

$$\bar{Q}_L(f) + \vartheta\langle\Phi,f\rangle^2 \;\ge\; \mu\|f\|^2 \qquad \text{for every } f \text{ in the form domain of sector } \varepsilon. \tag{7.1}$$

Consequently, if $\vartheta = 0$, *then* $\lambda_1^{\mathrm{ev}}(L) \ge \mu$ *(*$\varepsilon = +1$*) or* $\lambda_1^{\mathrm{odd}}(L) \ge \mu$ *(*$\varepsilon = -1$*); if* $\vartheta > 0$, *then* $\lambda_2^{\mathrm{ev}}(L) \ge \mu$.

Outline. By the form-core argument (coefficients of smooth f decay faster than any power of k_q, $A_{qr} - H(\kappa_q)\delta_{qr}$ is bounded, and $\bar{Q}_L$ is continuous in form norm) it suffices to prove (7.1) for rapidly decreasing coefficient vectors $x = (x_c, y, z)$ (core, I_P, far). (1) By (V2), completing the square in the core leaves the Schur complement $\mathcal{S}$ of the core; it inherits the displacement structure, $D_\lambda\mathcal{S} - \mathcal{S}D_\lambda = UV^{\mathsf{T}} - VU^{\mathsf{T}} + \vartheta(YX^{\mathsf{T}} - XY^{\mathsf{T}})$, of rank 2, or 4 with deflation, so its entries are explicit rational functions of λ_q evaluated through moments of the core solves. (2) By (V1) and Lemma 5.6, with moment and Abel-summation bounds for the far part of the core Schur complement, the far block satisfies $\mathcal{S}_{FF} \succeq \operatorname{diag}(t'_q)$, and (V3) makes this diagonal positive. (3) A second square completion removes the far block and leaves $S_M - R_M$ on I_P, with $0 \preceq R_M$ and $\|R_M\| \le r_M$ by a Hilbert–Schmidt bound. (4) $S_M \succeq S_{\mathrm{input}} - \varepsilon_{\mathrm{asm}} \succeq$

$(\delta_0 - \varepsilon_E - \varepsilon_{\mathrm{asm}})I$ by Lemma 7.4 and (V4), so (V5) gives $S_M - R_M \succeq 0$. For $\vartheta > 0$, every two-dimensional subspace of the form domain contains some $f \neq 0$ with $f \perp \Phi$, for which (7.1) reads $\bar{Q}_L(f) \geq \mu\|f\|^2$; min-max gives $\lambda_2^{\mathrm{ev}} \geq \mu$, as in Lemma 5.14. The full proof, with every series shown to converge absolutely, is [7, P3, P19–P23 and §8]. □

The explicit constants. The certifier evaluates the following formulas, each proved in the supplement [7]; `supplement/MAP.md` names the code line and the record field of each, and gate C-ind recomputes τ_{core}, τ_{pole}, t_N and r_M by independent code (Table 5). Write $k_q = q + \omega$, $\ell_* = k_{K-1}^2$, and let M^{-1} act on the core.

- *Generators* (P11–P12). Put $a = \mathsf{M}^{-1}u$, $b = \mathsf{M}^{-1}v$ (core parts of the generators) and, with deflation, $e = \mathsf{M}^{-1}\phi$, $\hat{e} = \mathsf{M}^{-1}D_\lambda\phi$; for $x \in \{u, v\}$ and $y \in \{a, b, e, \hat{e}\}$ let $R_{x,y}(z) = \sum_{i<K} x_i y_i/(z - \lambda_i)$. The transfer matrix $\mathcal{T}(z)$ has the rows $(1 - R_{v,a}, R_{u,a})$ and $(-R_{v,b}, 1+R_{u,b})$, and with deflation also $(R_{v,\hat{e}}, -R_{u,\hat{e}})$ and $(R_{v,e}, -R_{u,e})$, so $m = 2$ or 4; and $\mathcal{G}_q = \mathcal{T}(\lambda_q)(u_q, v_q)^{\mathsf{T}}$; the pairs are balanced by exact powers of two. The diagonal of the core Schur complement is $\mathcal{S}_{qq} = A_{qq} - \mu - Q_q$ with $Q_q = g_q^{\mathsf{T}}\mathcal{T}(\lambda_q)^{\mathsf{T}} J_\vartheta \mathcal{T}'(\lambda_q) g_q \geq 0$, proved by a limit of two virtual labels.
- *Moment bounds* (P14). With $m_r = \sum_i x_i y_i (\lambda_i/\ell_*)^r$, $A_f = \sum_i |x_i y_i|$ and $t_0 = \ell_*/z_0 < 1$, for $z \geq z_0$: $|zR(z)| \leq F_0$ and $|z^2 R'(z)| \leq F_1$, where

$$F_0 = \sum_{r<R} |m_r| t_0^r + \frac{A_f t_0^R}{1 - t_0}, \qquad F_1 = \sum_{r<R} (r+1)|m_r| t_0^r + \frac{A_f t_0^R (R + 1 - Rt_0)}{(1 - t_0)^2}.$$

  Then $L_{ab} := \sum_{\rho,\sigma} (|\delta_{\rho a}| + F_{0,\rho a}/z_0)\, |(J_b)_{\rho\sigma}|\, F_{1,\sigma b}$ at $z_0 = k_N^2$, where δ is the constant part of $\mathcal{T}$ (P15).
- *Oscillation constants* (P16–P17). $\bar{r} = \frac{2L}{\pi^2}\big[1 + \frac{e^{-L}}{1-e^{-4L}} + \frac{|\alpha_\varepsilon|}{1/4+\kappa_K^2}\big]$; with $\varphi_n = \pi \log n/L$, $V_h = \frac{1}{4} + \frac{2}{\pi^2}\sum w_n^2$, $E_1 = \sum_n w_n/|\sin\frac{\varphi_n}{2}|$, $E = \sum_{n\neq n'} \frac{w_n w_{n'}}{2|\sin\frac{\varphi_n - \varphi_{n'}}{2}|} + \sum_{n,n'} \frac{w_n w_{n'}}{2|\sin\frac{\varphi_n + \varphi_{n'}}{2}|}$ (every sine is certified nonzero), $E_h = \frac{2}{\pi}E_1 + \frac{4}{\pi^2}E$, $J_2(N) = \frac{V_h}{k_N - 1} + \frac{E_h}{k_N^2}$ and $J_4(N) = \frac{1}{3(k_N-1)^3}$.
- *Far block* (P18–P19). $\tau_N = \tau_{\mathrm{core}} + [\varepsilon = -1]\tau_{\mathrm{pole}}$ with $\tau_{\mathrm{core}} = L_{11}J_2 + (2\bar{r}L_{11} + L_{12} + L_{21})\sqrt{J_2 J_4} + (\bar{r}^2 L_{11} + \bar{r}(L_{12} + L_{21}) + L_{22})J_4$ and $\tau_{\mathrm{pole}} = \frac{2L^3}{\pi^4}\cosh^2\frac{L}{2}\, J_4$, at N. Then $t_q = H(\kappa_q) - \mu - s - \tau_N$, $t'_N = \lfloor t_N \rfloor_{40}$ and $t'_q = \max(t'_N, \lfloor t_q \rfloor_{40})$ for $N < q \leq M$, $t'_q = t'_M$ for $q > M$, where $\lfloor\cdot\rfloor_{40}$ rounds the lower end down to $2^{-40}\mathbb{Z}$.
- *Assembly debit* (P20–P21). On I_P, S_M has entries $h_i^{\mathsf{T}}\mathcal{J}h_j/(\lambda_i - \lambda_j)$ with $h_i = (\mathcal{G}_i, w_i)$, $w_i = J_b Z(i) J_b^{\mathsf{T}}\mathcal{G}_i$, $Z(i) = \sum_{q\in[N,M)} \mathcal{G}_q\mathcal{G}_q^{\mathsf{T}}/(t'_q(\lambda_i - \lambda_q))$ and $\mathcal{J} = \left(\begin{smallmatrix} J_b & -I \\ I & 0 \end{smallmatrix}\right)$. With $H = (1 + \log(N - K))/K$, e_h and e_d the radius plus rounding of the rows and diagonal, $\varepsilon_{\mathrm{asm}} = e_d + H\sum_{a,b} |\mathcal{J}_{ab}|\big(e_h(|\hat{h}|_a + |\hat{h}|_b) + e_h^2\big)$.
- *Omitted tail* (P22). With $\varrho = k_{N-1}/k_M$, $S_a \geq \sum_{i\in I_P} \mathcal{G}_{ia}^2$, $A_\rho = |\delta_{\rho 1}| + F_{0,\rho 1}/k_M^2$, $C_\rho = |\delta_{\rho 2}| + F_{0,\rho 2}/k_M^2$ (at $z_0 = k_M^2$) and $\beta_\rho = A_\rho\sqrt{J_2(M)} + (A_\rho\bar{r} + C_\rho)\sqrt{J_4(M)}$: $r_M = \big[\sum_{a,b} |(J_b)_{ab}|\sqrt{S_a}\,\beta_b\big]^2/\big(t'_M(1 - \varrho^2)^2\big)$.

Every quantity is an Arb upper end or an exact rational. δ_0 is the smallest multiple of 2^{-8} that is at least $r_M + \varepsilon_{\mathrm{asm}} + 2^{-10}$.

(V1) is certified by Elston's weighted Schur test (the routine behind (H1) of Proposition 5.13) on $L^2(-b, b)$ with $b = \lceil 2^{24}L\rceil/2^{24}$; we call this *stage S*. (V2) is certified as in Proposition 5.13(c): an integer Cholesky proposal and an exactly computed integer residual give $\lambda_{\min}(\mathsf{M}) \geq m_{\mathrm{core}} > 0$, and the core solves are enclosed through $\|\mathsf{M}^{-1}\| \leq 1/m_{\mathrm{core}}$. All other quantities are Arb balls or exact rationals; floating-point values (Galerkin eigenvalues, ϕ, proposals of pivots and

multipliers) only choose parameters. The certified lower bound of a run is the exact dyadic μ itself.

Lemma 7.4 (Exact structured LDL^{T}; Lemma SR)**.** *Let $\lambda_1, \dots, \lambda_n$ be distinct rationals, $\mathcal{J}$ a skew-symmetric integer $D \times D$ matrix with $|\mathcal{J}|_1 = \sum_{a,b} |\mathcal{J}_{ab}|$, $h_j \in (2^{-P}\mathbb{Z})^D$ and $d_j \in 2^{-P}\mathbb{Z}$, and let $\hat{S}_{jk} = h_j^{\mathsf{T}} \mathcal{J} h_k/(\lambda_j - \lambda_k)$ for $j \neq k$ and $\hat{S}_{jj} = d_j$. For $i = 1, \dots, n$, with the current h and d: require the pivot $p_i := d_i > 0$; for $j > i$ put $\nu_j = h_j^{\mathsf{T}} \mathcal{J} h_i$, choose any rational ℓ_j, and replace h_j by $\lfloor h_j - \ell_j h_i \rfloor_P$ and d_j by $\lfloor d_j - p_i \ell_j^2 \rfloor_P$, where $\lfloor \cdot \rfloor_P$ rounds down to $2^{-P}\mathbb{Z}$; record $e_i = \max_j |\nu_j - p_i \ell_j (\lambda_j - \lambda_i)|$, $\bar{\ell}_i = \max_j |\ell_j|$ and $G_i = \max_j \|h_j - \ell_j h_i\|_\infty$. If every pivot is positive, then $\lambda_{\min}(\hat{S}) \geq -\varepsilon_E$ with*

$$\varepsilon_E = \sum_{i=1}^{n} \Big[2^{-P} + \big(e_i + 2\bar{\ell}_i e_i + |\mathcal{J}|_1 2^{-P} (2G_i + 2^{-P}) \big) H_S \Big], \qquad H_S \geq \max_j \sum_{k \neq j} |\lambda_j - \lambda_k|^{-1}.$$

Proof. Let $S^{(i)}$ be the current matrix on the indices $\geq i$, $\ell^{(i)} = 1_i + \sum_{j>i} \ell_j 1_j$ and $E_i = S^{(i)} - p_i \ell^{(i)} \ell^{(i)\mathsf{T}} - S^{(i+1)}$ (zero-extended). Then $\hat{S} = \sum_i p_i \ell^{(i)} \ell^{(i)\mathsf{T}} + \sum_i E_i$ exactly, and the first sum is $\succeq 0$. Using $h_i^{\mathsf{T}} \mathcal{J} h_i = 0$ one finds $(E_i)_{ji} = (\nu_j - p_i \ell_j (\lambda_j - \lambda_i))/(\lambda_j - \lambda_i)$, $(E_i)_{jj} \in [0, 2^{-P})$, and $(E_i)_{jk} = \nu^E_{jk}/(\lambda_j - \lambda_k)$ for $j \neq k > i$ with $|\nu^E_{jk}| \leq 2\bar{\ell}_i e_i + |\mathcal{J}|_1 2^{-P}(2G_i + 2^{-P})$. A symmetric matrix has spectral norm at most its largest absolute row sum, so $\|E_i\| \leq 2^{-P} + (e_i + 2\bar{\ell}_i e_i + |\mathcal{J}|_1 2^{-P}(2G_i + 2^{-P}))H_S$. □

In the runs the rows are $h_i = (\mathcal{G}_i, w_i)$ with $D = 4$ ($\vartheta = 0$) or $D = 8$ ($\vartheta > 0$), $P = 96$, the multipliers are P-bit dyadics, and every quantity is an integer after scaling by 2^P. The cost is $O(n^2 D)$ integer operations and $O(nD)$ memory, with $n = N - K$ up to about $6 \cdot 10^4$; ε_E is an exact rational, and gate R6 requires $\varepsilon_E \leq 2^{-12}$. The step test (3.1) then compares the certified μ's with the upper bounds $U^{\mathrm{ev}}(L_i)$ of §8.

7.4. **Gates.** A run counts only if every hard gate holds; a false hard gate or a crash voids the run and stops the grid. Table 5 lists the gates.

7.5. **Grid, pre-registration and step test.** *Pre-registration.* The protocol (windows, parameters, rungs, gates, success and kill criteria, a partial-outcome rule and float predictions) was frozen by a stamp holding the hashes of every document and executed code file before any grid computation; changes were allowed only as entries of an append-only, hash-chained amendment log written before any result they could affect. Every run record carries the stamp hash, the hashes of the archived code it executed, and a pin of the amendment log.

Windows. 53 windows were registered: $L_0 = 1.4035$ (the end of Piece III, where only U^{ev} is needed), Elston's windows 1.4255, 1.447, 1.468, 1.4885 and 1.5 as exact decimals (for the cross-check), $L = 2$, and otherwise dyadics $m/2^{14}$, placed, from float forecasts, as few as possible between the anchors subject to a forecast $\frac{2}{3} U^{\mathrm{odd}}(L_{i+1})/U^{\mathrm{ev}}(L_i) \geq 2$; every window satisfies $|2L - \log n| > 2^{-20}$ for all prime powers n. One refinement window was added under the registered rule (§7.6), giving the 54 windows and 53 steps of Table 6.

Upper bounds. $U^{\mathrm{ev}}(L_i)$ and $U^{\mathrm{odd}}(L_i)$ are Rayleigh-quotient upper bounds from §8, with $K = 640$, 768 or 896 modes and 3000–6000 bits for the new windows, using the shipped code of `code/upper/`; the bounds at 1.4035, 1.4255, 1.447, 1.468, 1.4885, 1.5 and 2 are earlier certified files. U^{odd} serves only to choose μ and for the guard N2.

Certifier runs. $K = 512$; N is the smallest multiple of 1024 with $t_N \geq t_{\min}$ ($t_{\min} = 1$ for λ_1^{odd}, $\frac{1}{2}$ for λ_2^{ev}), and $M = 4N$ (or $8N$). Each window has a ladder of at most three runs per kind, stopping at the first pass: $\mu \in \{\frac{2}{3}, \frac{1}{2}, \frac{1}{3}\} \cdot U^{\mathrm{odd}}(L)$ for λ_1^{odd}, and $\mu \in \{\frac{1}{4}\hat{\lambda}_2, 2^{-10}\hat{\lambda}_2, 2U^{\mathrm{ev}}(L_{i-1})\}$

TABLE 5. Gates of Piece IV. Upper part: hard gates in every certifier run (all 15 are recorded in each run file). Lower part: gates per grid.

gate	content
V2–V5	the hypotheses of Theorem 7.3, decided in Arb or exact arithmetic
G-S	$s(L)$ exceeds the float Ritz values of the prime comb (1600 modes, both parities) and stage S's own float top eigenvalue; catches gross errors in s only
G0-mini	closed-form entries against (2.2) at 4 entries, inside the run
G0i	8 entries of the run's sector recomputed from (2.2) by independent code (own prime-power list and weights, mpmath at 30 digits), tolerance 10^{-20}
G5	core solves (own Cholesky) and the diagonal correction at 32 indices against a direct Arb solve; S_{input} entries rebuilt with the labels that Lemma 7.4 reads, against a direct dense evaluation
G5′	every generator and diagonal on $[K, M)$ by a second path (direct sums for $q < 2K$, moment series beyond)
C-τ, C-$\bar{r}$, C-J_2, C-r_M	computed partial sums and maxima do not exceed the analytic bounds that enter τ_N and r_M
C-ind	τ_N, the pole part, r_M and t_N recomputed by independent mpmath code from the run's core solves
R6	P-bit dyadic multipliers and $\varepsilon_E \le 2^{-12}$
P0	exact rational checks of the displacement, diagonal-correction, partial-fraction and Schur-structure identities and of Lemma 7.4; the C kernel of Lemma 7.4 equals the Python reference
T15	toolchain reproduction: two runs at $L = 1.5$ reproduce earlier records in every certified field
G0	full Arb comparison of closed forms and (2.2), both sectors, at every window
G2	6 negative controls ($\mu = 1.01\,U^{\sigma}(L)$ at three windows, both sectors) must fail
G4	15 deliberate code mutations at $L = 1.5$ must each be caught
N2	a pass with μ above the reference ($U^{\mathrm{odd}}(L)$, or the Galerkin $\hat{\lambda}_2$ of the core) stops the grid
F6	the gluing accepts only passing records with the right window, sector, no mutation flag and the grid stamp

for λ_2^{ev}, where $\hat{\lambda}_2$ is the float second Galerkin eigenvalue of the core. The even lowest eigenvalue is not certified from below; only U^{ev} enters the step test.

Step test and gluing. The gluing script takes, per window and kind, the first passing record in sorted order, applies F6, and checks (3.1) in Arb, step by step, over the registered window list.

7.6. **The registered failure and rule E.** *What failed.* At $L_{44} = 1.92364501953125$ all three λ_2^{ev} rungs failed hypothesis (V4) only, every other hard gate being true: each time Lemma 7.4 met a negative pivot at the last of its 37376 labels ($N = 37888$). The odd sector passed there. The registered refinement inserted $L' = 31443/2^{14} = 1.91912841796875$; the step $[1.91455078125, L']$ passed, but $[L', 1.92364501953125]$ still had no λ_2^{ev} bound, because under the registered reuse rule the ladder at 1.92364501953125 kept its three failed records. Only one refinement level per step was allowed, so the registered kill criterion applied: *the grid is not viable as registered.* The registered partial outcome, the longest passing run of steps from L_0, is $[1.4035, 1.91912841796875]$: 43 steps, smallest ratio 2.098, later confirmed by a scored partial gluing.

Diagnosis (post hoc, not scored). The failure is a boundary effect of the cut N. The last row $q = N - 1$ of I_P couples to the first far mode through an entry of order 1 that depends on (L, N) but not on μ, while the far block is bounded below only by $t'_N \approx 0.6$ (its true diagonal

is about 9); the resulting loss makes S_{input} indefinite even as $\mu \to 0$. A diagnostic run at $N + 1024 = 38912$ passed. A failure of this kind cannot produce a false pass: replacing the far block by $\mathrm{diag}(t'_q) \preceq \mathcal{S}_{FF}$ only lowers S_M. The window 1.8958740234375 showed the same pattern on its first rung and passed on its second rung ($2^{-10}\hat{\lambda}_2$, with $N = 31744$).

Rule E (registered post hoc). If a window's λ_2^{ev} ladder ends without a pass and all three rungs failed at (V4) only, run one further rung $\mu = \frac{1}{4}\hat{\lambda}_2$ with $t_{\min}$ raised to $t_N + 2^{-10}$ of the first rung, which moves N to the next multiple of 1024. The rule changes one argument that only selects N; the certifier code, every gate and every tolerance are unchanged, and all are recomputed at the new N. Since N is a free parameter of Theorem 7.3, the rule cannot weaken a certificate; it is nevertheless a choice made after seeing a failure, and it was registered with its own stamp, scored runs, negative control and fallback before the scored run. It fires only at 1.92364501953125 ($N = 37888 \to 38912$).

Scored outcome. The rule-E run passed with all 15 gates true: $\lambda_2^{\mathrm{ev}}(1.92364501953125) \geq 4.571 \cdot 10^{-232}$, against $U^{\mathrm{ev}}(L') \leq 2.748 \cdot 10^{-239}$ (the step $[L', 1.92364501953125]$ has ratio 448.6 and is odd-limited). The full gluing then passed all 53 steps. The negative control (same window and parameters, $\mu = 1.01\,\hat{\lambda}_2$) failed as required, but at (V2), in the core, rather than at (V4) at $N = 38912$ as predicted; it is therefore a weaker test of the Lemma SR path at the new N than planned. That this path detects failures rests on the 6 grid controls, which all failed at (V4), and on the four failed grid records, which all failed at the last label. The registered fallback (substituting a λ_2^{ev} bound from a larger window via Lemma 3.1) was not triggered, because rule E passed.

Remark 7.5 (Rule E is not needed). The fallback route works on its own. By Lemma 3.1,

$$\lambda_2^{\mathrm{ev}}(1.92364501953125) \;\geq\; \lambda_2^{\mathrm{ev}}(1.9326171875) \;\geq\; 1.3019115357395 \cdot 10^{-236},$$

where the last bound is the registered grid record at 1.9326171875 (rung $\frac{1}{4}\hat{\lambda}_2$, $N = 39936$, `data/piece4/glue/g_1.9326171875_l2.json`). It exceeds the odd bound $\lambda_1^{\mathrm{odd}}(1.92364501953125) \geq 1.2323704419343 \cdot 10^{-236}$, so the step $[1.91912841796875, 1.92364501953125]$ satisfies (3.1) with the same, odd-limited ratio 448.6, and no other step changes. The gluing script with `--l2-from 1.92364501953125=1.9326171875`, which reads that record in place of the rule-E record, passes all 53 comparisons with the same minimum ratio 2.098; `check_theorem.sh` runs this route too. The substitution was written down after the failure, as the fallback of the rule-E registration, but it changes no parameter and uses only records of the registered grid. A reader who rejects rule E therefore still obtains Theorem 4.1 on $(0, 2]$. Earlier drafts said that without rule E the theorem covers only $(0, 1.91912841796875]$; that was too strong, as the second AI review (§10.3) pointed out. What remains true is that grid G failed as a pre-registered experiment.

7.7. **Results.** Table 6 lists the certificate. All 53 steps satisfy (3.1), and the odd sector binds at every step. The smallest ratio is 2.098, on $[1.5, 1.5174560546875]$; on $[1.4035, 1.5]$ the ratios are about 50–120, because Elston's windows there, with the dyadic points between them, are spaced more closely than needed. At $L = 2$, $\lambda_1^{\mathrm{odd}}(2) \geq 2.158 \cdot 10^{-278}$ and $\lambda_2^{\mathrm{ev}}(2) \geq 1.365 \cdot 10^{-273}$, against $\lambda_1^{\mathrm{ev}}(2) \leq U^{\mathrm{ev}}(2) \leq 1.847 \cdot 10^{-283}$ (ratio $1.168 \cdot 10^5$). By Corollary 3.2 this proves item (4) of Theorem 4.1. □

Cost. The registered grid ran 372 jobs (certifier runs, upper bounds, stage S, Ritz guards, gates and controls) in 87.4 core-hours on one 64-vCPU virtual machine with 60 GB of memory (7.5 h wall); the upper bounds took two thirds of it. Each certifier run is single-core; the glued runs took 172–2630 s each (field `seconds`), with peak memory below 3.5 GB. The rule-E stage took 0.34 core-hours.

TABLE 6. Piece IV certificate, generated from the run records by `paper/gen_tables.py` (no number typed by hand). $U^{\mathrm{ev}}(L_i) \geq \lambda_1^{\mathrm{ev}}(L_i)$ is the smallest rigorous Galerkin upper bound (§8); ℓ^{odd} and ℓ_2 are the certified lower bounds for λ_1^{odd} and λ_2^{ev} (Theorem 7.3); the step ratio in row i is $\min(\ell^{\mathrm{odd}}, \ell_2)(L_i)/U^{\mathrm{ev}}(L_{i-1})$. Lower bounds are rounded down, upper bounds up and ratios down, to four significant digits. N^{odd}, N_2: cut of the far block in the two runs ($K = 512$). Row 0 is the end of Piece III; $U^{\mathrm{ev}}(2)$ enters no step. Every odd bound is on the rung $\frac{2}{3}U^{\mathrm{odd}}$ and every λ_2^{ev} bound on $\frac{1}{4}\hat{\lambda}_2$, except † (rung $2^{-10}\hat{\lambda}_2$) and ‡ (rule E, §7.6). Row 43 is the refinement window $L' = 31443/2^{14}$.

i	L_i	$U^{\mathrm{ev}}(L_i)$	$\lambda_1^{\mathrm{odd}} \geq$	$\lambda_2^{\mathrm{ev}} \geq$	ratio $\geq$	N^{odd}	N_2
0	1.4035	2.493e-78	—	—	—	—	—
1	1.41473388671875	2.433e-80	2.726e-76	9.580e-73	109.3	3072	3072
2	1.4255	2.371e-82	2.762e-78	1.068e-74	113.5	4096	3072
3	1.4366455078125	2.146e-84	2.631e-80	9.514e-77	111.0	4096	3072
4	1.447	2.107e-86	2.512e-82	1.102e-78	117.1	4096	4096
5	1.4576416015625	1.836e-88	2.389e-84	9.629e-81	113.4	4096	4096
6	1.468	1.740e-90	2.165e-86	9.055e-83	118.0	4096	4096
7	1.478271484375	1.250e-92	1.891e-88	9.330e-85	108.6	5120	4096
8	1.4885	1.084e-94	1.516e-90	6.790e-87	121.3	5120	4096
9	1.5	3.852e-97	5.478e-93	2.736e-89	50.56	5120	4096
10	1.5174560546875	5.172e-101	8.081e-97	4.813e-93	2.098	6144	5120
11	1.53466796875	6.990e-105	1.193e-100	7.206e-97	2.308	6144	5120
12	1.5511474609375	9.681e-109	1.641e-104	1.022e-100	2.349	6144	6144
13	1.5672607421875	1.084e-112	2.224e-108	1.462e-104	2.298	7168	6144
14	1.58306884765625	1.178e-116	2.614e-112	1.838e-108	2.413	7168	7168
15	1.5986328125	1.119e-120	2.708e-116	2.151e-112	2.300	8192	7168
16	1.61376953125	1.052e-124	2.645e-120	2.316e-116	2.365	9216	8192
17	1.62847900390625	8.653e-129	2.413e-124	2.356e-120	2.295	9216	8192
18	1.642822265625	7.442e-133	2.033e-128	2.083e-124	2.350	10240	9216
19	1.65692138671875	6.307e-137	1.805e-132	1.847e-128	2.426	10240	9216
20	1.6705322265625	5.011e-141	1.456e-136	1.617e-132	2.309	11264	10240
21	1.6839599609375	3.743e-145	1.214e-140	1.393e-136	2.424	12288	11264
22	1.69708251953125	2.457e-149	8.640e-145	1.052e-140	2.308	13312	11264
23	1.71002197265625	1.555e-153	5.648e-149	7.392e-145	2.299	14336	12288
24	1.7225341796875	9.995e-158	3.691e-153	5.609e-149	2.375	15360	13312
25	1.73504638671875	5.618e-162	2.316e-157	3.362e-153	2.317	15360	14336
26	1.74713134765625	3.302e-166	1.319e-161	1.882e-157	2.349	16384	15360
27	1.759033203125	1.664e-170	7.707e-166	1.163e-161	2.334	17408	15360
28	1.770751953125	7.915e-175	3.943e-170	6.767e-166	2.370	19456	16384
29	1.78216552734375	3.978e-179	1.914e-174	3.517e-170	2.419	20480	17408
30	1.7933349609375	1.967e-183	9.546e-179	1.942e-174	2.400	21504	18432
31	1.80438232421875	8.707e-188	4.544e-183	8.954e-179	2.311	22528	19456
32	1.81524658203125	3.824e-192	2.097e-187	4.230e-183	2.408	23552	21504
33	1.825927734375	1.407e-196	8.810e-192	2.025e-187	2.304	25600	22528
34	1.83642578125	5.028e-201	3.216e-196	8.125e-192	2.286	26624	23552
35	1.8468017578125	1.914e-205	1.155e-200	2.810e-196	2.299	27648	24576
36	1.85687255859375	6.975e-210	4.585e-205	1.219e-200	2.396	30720	26624
37	1.86688232421875	2.278e-214	1.611e-209	4.568e-205	2.310	31744	27648
38	1.87664794921875	6.803e-219	5.014e-214	1.579e-209	2.201	32768	28672
39	1.8863525390625	2.203e-223	1.657e-218	5.264e-214	2.437	34816	30720
40	1.8958740234375	6.749e-228	5.479e-223	6.678e-221†	2.487	36864	31744
41	1.9052734375	1.911e-232	1.545e-227	5.446e-223	2.289	38912	33792
42	1.91455078125	5.159e-237	4.422e-232	1.661e-227	2.314	40960	35840
43	1.91912841796875	2.748e-239	2.424e-234	8.621e-230	469.9	41984	36864
44	1.92364501953125	1.421e-241	1.232e-236	4.571e-232‡	448.6	43008	38912
45	1.9326171875	3.655e-246	3.391e-241	1.301e-236	2.386	46080	39936
46	1.94146728515625	8.725e-251	8.111e-246	3.477e-241	2.219	48128	41984
47	1.95013427734375	1.947e-255	1.958e-250	9.753e-246	2.244	51200	44032
48	1.958740234375	4.587e-260	4.498e-255	2.121e-250	2.310	54272	46080
49	1.96722412109375	1.047e-264	1.105e-259	5.273e-255	2.411	56320	48128
50	1.9755859375	2.270e-269	2.398e-264	1.322e-259	2.291	59392	51200
51	1.98382568359375	4.665e-274	5.124e-269	3.009e-264	2.258	62464	53248
52	1.99200439453125	9.233e-279	1.074e-273	6.330e-269	2.303	66560	56320

7.8. **Cross-check with Piece III.** At the four windows that both grids share past 1.4035, the two routes give certified lower bounds for the same eigenvalues by different reductions, matrices and factorisations (Table 7). The ratios ours/Elston are $0.7639 - -0.7751$ for λ_1^{odd} and $0.3137 - -0.3221$ for λ_2^{ev}. This is as designed: Elston's odd bounds lie within about 10^{-6} of his own Galerkin values, while ours sit on the rungs $\frac{2}{3}U^{\mathrm{odd}}$ and $\frac{1}{4}\hat{\lambda}_2$, and U^{odd} ($K = 640$) is about 15% above λ_1^{odd} there. Elston's odd bounds lie below our rigorous U^{odd} (checked by `paper/gen_tables.py`). Our runs take minutes on one core, Elston's several hours with 11–29 worker processes. On $[1.4035, 1.4885]$ the step test passes independently in both routes (Tables 4 and 6).

TABLE 7. Cross-check of Piece IV ("ours") with Piece III (Elston's certifier with our deflation) at the shared windows; lower bounds rounded down, ratios floored, times in seconds (field `seconds`; ours single-core). Generated by `paper/gen_tables.py`.

	$\lambda_1^{\mathrm{odd}} \geq$			$\lambda_2^{\mathrm{ev}} \geq$			time, odd, λ_2	
L	ours	Elston	ratio	ours	Elston	ratio	ours	Elston
1.4255	2.762e-78	3.564e-78	0.7751	1.068e-74	3.366e-74	0.3174	202, 230	13217, 18715
1.447	2.512e-82	3.269e-82	0.7683	1.102e-78	3.420e-78	0.3221	224, 244	20970, 24322
1.468	2.165e-86	2.834e-86	0.7639	9.055e-83	2.886e-82	0.3137	215, 265	25765, 31456
1.4885	1.516e-90	1.964e-90	0.7720	6.790e-87	2.141e-86	0.3170	172, 213	44638, 40211

7.9. **Imported trust.** Piece IV is conditional on the following, none of which was re-verified by the audits of §10.3:

(1) *Stage S*, the value $s(L)$ of (V1) at every window: Elston's weighted Schur test run through our wrapper. Gate G-S catches only errors larger than the gap between s and the Ritz values (about 0.004–0.006); an s that is too small would be invisible to every other gate.
(2) *Elston's integer Cholesky and residual code*, which decides (V2) in every run; it was audited independently only at $L = 1.19$ [16].
(3) *The upper bounds* $U(L)$ of §8 and their outward rounding.
(4) *Our certifier* (Python with a C kernel for Lemma 7.4) beyond its records, and the proof document (now public as [7]); they rest on AI reviews and audits, the P0 identities and the 15 caught mutations.
(5) *Arb/FLINT* (python-flint 0.9.0), including the complex digamma and polygamma functions, and mpmath for the independent gates.
(6) *This paper*: the operator setup of §2 (closability, compact resolvent, min-max), Lemmas 5.6, 3.1 and 5.14, Corollary 3.2, and Proposition 7.2, which is checked only indirectly through G0 and G0i.
(7) *Pieces I–III* on $(0, 1.4035]$, with their own caveats (§11).
(8) *The compute host* reported file contents and times honestly; every mirrored record matches it by SHA-256.

8. Rigorous upper bounds for larger windows

This section gives *upper bounds only*; it makes no lower-bound, positivity or simplicity statement. The same code gives the bounds $U(L_i)$ of Pieces III and IV (Tables 4 and 6).

Galerkin matrix. Let $e_k(y) = \sqrt{(2k+1)/(2L)}P_k(y/L)$ with $k = 2m+p$, $m < K$, $p \in \{0,1\}$ the parity. With $r = x/L \in [0,2]$, the symmetrised autocorrelation of e_a, e_b is an L-independent polynomial in r: up to the normalisation $\sqrt{(2a+1)(2b+1)}/2$ it equals $C_{ab}(r) = \int_{-1}^{1-r} P_a(t)P_b(t+r)\,dt$ (for $a+b$ even $C_{ab} = C_{ba}$). The archimedean weight $k_L(r) = 2Le^{-Lr/2}/(1-e^{-2Lr})$ behaves like $1/r$ at $r = 0$ and is not integrable, so it enters only through a regularised moment. In (2.2), with $x = Lr$,

$$\int_0^{2L} \frac{2[e^{-2x}g(0) - e^{-x/2}g(x)]}{1-e^{-2x}}\,dx = \int_0^2 k_L(r)\big(g(0) - g(Lr)\big)\,dr + d_L\,g(0),$$

$$d_L = \int_0^2 \frac{2L(e^{-2Lr} - e^{-Lr/2})}{1-e^{-2Lr}}\,dr,$$

where both integrands are analytic on $[0,2]$ (the singularity at $r = 0$ is removable: $g(0) - g(Lr) = O(r)$, and the integrand of d_L tends to $-\frac{3}{2}L$). Since $g(0) = \|f\|^2$, the term $d_L g(0)$ and the constant terms of (2.2) form the diagonal correction $(d_L + c_L)I$ with $c_L = -(\log\pi + \gamma + \log(1-e^{-4L}))$. Let m be the linear functional $P_j(r-1) \mapsto m_j$ on polynomials. The Galerkin matrix is $Q_K = \big(\frac{1}{2}\sqrt{(2a+1)(2b+1)}\,\mathsf{m}(C_{ab})\big)_{a,b} + (d_L + c_L)I$, with the moments

$$m_j = \int_0^2 4L\cosh\tfrac{Lr}{2}\,P_j(r-1)\,dr + \int_0^2 k_L(r)\big(P_j(-1) - P_j(r-1)\big)\,dr - \sum_{\log n < 2L} \frac{2\Lambda(n)}{\sqrt{n}} P_j\Big(\frac{\log n}{L} - 1\Big).$$

The values $\mathsf{m}(C_{ab})$ follow from the moments by the convolution recurrence [19], applied through the linear functional m,

$$C_{a+1,b} = C_{a-1,b} + \frac{2a+1}{2b+1}\big(C_{a,b-1} - C_{a,b+1}\big),$$

with exact seeds C_{0b}, C_{1b} expressed through $\int_u^1 P_k$. The recurrence is run only on $b \geq a$, where it is forward stable; the cost is $O(K^2)$ instead of $O(K^4)$ for direct quadrature.

Proposition 8.1 (Rigorous moments)**.** *The two integrals in m_j and d_L are computed with M_q-point Gauss–Legendre in $u = r-1 \in [-1,1]$, and the truncation error of each is bounded by Lemma 5.8. The archimedean integrand is written as $h_L(r)\,\frac{P_j(-1)-P_j(r-1)}{r}$ with $h_L(r) = rk_L(r) = 2Lr\,e^{-Lr/2}/(1-e^{-2Lr})$; both factors are analytic inside E_ρ (the first has poles only at $r = i\pi m/L$, $m \neq 0$, which are checked to lie outside E_ρ; the second is a polynomial), so by the maximum principle it is bounded on E_ρ by $\max_{\partial E_\rho}|h_L| \cdot (1+\rho^j)/\min_{\partial E_\rho}|r|$, using $|P_j| \leq \rho^j$ on E_ρ. The maxima of $4L\cosh(Lr/2)$, of h_L and of the integrand of d_L on ∂E_ρ are enclosed by covering it with 2000 complex balls. The Legendre values at the nodes (and at $\log n/L - 1$) are computed as follows: the Chebyshev recurrence $T_j = 2xT_{j-1} - T_{j-2}$ is run on exact midpoints, re-centred at every step, with the per-step rounding asserted to be at most $\delta = 2^{8-p}$ at precision p; the accumulated error of $\sum_n c_n T_j(x_n)$ is then at most $\sum_n |c_n|\,j^2(\delta/2 + r_x)$, because the error of the recurrence is $\sum_i \delta_i U_{j-i-1}(x)$ with $|U_m| \leq m+1$ on $[-1,1]$, and a node radius r_x contributes at most $j^2 r_x$ by Markov's inequality $|T_j'| \leq j^2$. This bound is added as a ball, and $P_n = \sum_m \lambda_m\lambda_{n-m}(2 - [n = 2m])T_{n-2m}$ with $\lambda_m = \binom{2m}{m}4^{-m}$, a map with nonnegative coefficients and row sums 1, does not amplify it. Hence every entry of Q_K is enclosed in a ball, and for every exact vector v the upper endpoint of $v^T Q_K v / v^T v$ is an upper bound for $\lambda_1^\sigma(L)$ in the sector of v.*

The last statement holds because v defines a polynomial trial function in the form domain (§5.2). The vector v comes from inverse iteration on midpoints and carries no rigour; the reported value is outward-rounded.

TABLE 8. Rigorous upper bounds for the lowest eigenvalue in each sector. K = number of Legendre modes per sector; "—" = no rigorous run. The Piece III windows are in Table 4 ($K = 640$, 3000 bits) and the Piece IV windows in Table 6 ($K = 640$–1024, 2400–6000 bits). Since release 2.0 the even bound at $L = 1.5$ enters the proof (Piece IV; Remark 8.2). The even values at $L = 2$ and 2.25 are explained in Remark 8.2(a).

L	$\lambda_1^{\mathrm{ev}}(L) \le$	$\lambda_1^{\mathrm{odd}}(L) \le$	K, bits (even; odd)
1.5	3.85105826960019e-97	8.21780562022764e-93	384, 2400; 512, 4000
1.75	—	2.06808416452444e-162	—; 512, 4000
2.0	1.84681871820109e-283	3.23821041502563e-278	1024, 6000; 1024, 6000
2.25	9.03204367910e-476	4.64181809939520e-470	2048, 8000; 1792, 8000

At $L = 2$ our even bound is below Zhu's certified $3.19 \cdot 10^{-283}$ [Table 3]; certified upper bounds for $L \le 2$ are not new. The window $L = 2.25$ lies outside the range of [38]. There ($\mu = e^{2L} \approx 90.0$) the explicit analytic bounds we know of are larger: about $e^{-1052.9}$ from [27, Thm. K(a)] (in his normalisation) and about $e^{-504.0}$ from [9, Thm. 2.1], against $e^{-1093.8}$ for our even bound; [32, Thm. 1] assumes RH and has a non-explicit $O(L^2)$ term, and [10, Cor. 2] is unconditional and has the sharp scale but non-explicit constants C_n, L_n, so neither gives a value. The values of Table 8 at $L = 2.25$ are therefore the sharpest explicit upper bounds there that we know of; the decay scale $-\ln \lambda \approx 4\pi e^{2L}$ is not ours [11, 27, 31, 32]. Non-rigorous Galerkin values continue to decrease slowly with K (at $L = 2.25$: $9.894 \cdot 10^{-476}$, $9.337 \cdot 10^{-476}$, $9.032 \cdot 10^{-476}$ for $K = 1536, 1792, 2048$); we do not have a bound on the omitted-mode tail, so these upper bounds may be far from sharp in relative terms.

Remark 8.2 (Decimal export and quadrature exponent). (a) *Decimal export.* Every upper bound is exported as a 15-digit decimal string with `str(15, radius=False)`. That conversion rounds to nearest, not in a fixed direction: its result can lie below its input, by at most one unit of the 15th digit (a relative 10^{-14}; on 20 000 random inputs at 3000 bits the largest relative error was $4.9 \cdot 10^{-15}$). `fr_sweep.py` as shipped therefore raises the upper end of the Rayleigh-quotient ball by a relative 10^{-13} before the conversion, so the exported string is above the true upper end, since in every shipped record the radius is below 10^{-556} times the value. Every shipped upper bound carries this raise except the original even runs at $L = 2$ and $L = 2.25$ (finished 28 and 30 October), which exported the upper end without it (their fields `lam0_mid` and `lam0_upper` print the same string), so their last digit may have been rounded down, and they are not used as printed. For $L = 2$, Table 8 gives the rerun of 31 October 2026 with the shipped code ($K = 1024$, 6000 bits, corrected exponent, with the raise; `data/piece4/glue/upper_rerun_L2.jsonl`), which is also the $U^{\mathrm{ev}}(2)$ of Table 6; the original printed $1.84681871820091 \cdot 10^{-283}$ (`data/upper/rig_L2.jsonl`). For $L = 2.25$ there is no rerun (it took 64 000 s). The record (`data/upper/rig_L225.jsonl`) prints $9.03204367909380 \cdot 10^{-476}$ with radius $1.07 \cdot 10^{-2022}$; the true upper end is within a relative 10^{-14} of the printed value, and Table 8 gives that value rounded up to 12 digits, $9.03204367910 \cdot 10^{-476}$, a relative $6 \cdot 10^{-12}$ higher. That margin also covers the quadrature question of (b), since the radius is a relative 10^{-1546} of the value. Neither value enters a step of the proof.

(b) *Quadrature exponent.* The original $L = 2$ even run was made before the exponent of Lemma 5.8 was corrected from ρ^{-2n} to ρ^{-2n+2} in this code path ($\rho \le 3$ there); the rerun of (a) uses the corrected exponent. The bounds U of Table 4 were computed on 30 October 2026 with `fr_sweep.py` as shipped in `code/upper/`: the exported value is the upper end of the Rayleigh-quotient ball raised by a relative 10^{-13} before rounding to 15 digits, while every ball

radius is below 10^{-600} times the value, so a factor $\rho^2 \le 9$ on the quadrature part of the radius cannot reach that margin, whichever exponent was in use. The working directory was not under version control, so the revision of `fast_recurrence.py` used for each run is not recorded. Re-running $U(1.19)$ with the shipped code on 31 October 2026 reproduced every exported digit and the radius (`data/piece3/rerun_U1.19.jsonl`). For the other rows of Table 8 (28–30 October) we have not confirmed which exponent was used. One of them, $U^{\mathrm{ev}}(1.5)$ ($K = 384$), enters Piece IV; its recorded radius is below 10^{-550} times the value, while the exported value exceeds the midpoint by a relative 10^{-13}, so the argument above applies to it as well. The other rows enter no proof. The new bounds $U(L_i)$ of Piece IV were computed on 31 October 2026 with the shipped `code/upper/` (hashes recorded in the run-time stamp); each exported upper end exceeds its midpoint by at most a relative $1.1 \cdot 10^{-13}$, and every radius is far smaller.

9. A numerical observation: prime-power singular points

Everything in this section is numerical and non-rigorous. It concerns finite-dimensional approximants of the even minimiser; whether the limiting eigenfunction has the same features is not established. That near-null eigenvectors have weak singularities at $\log n$ and $L - \log n$, and that meshes should be cut there, is due to Eberwein [14, §13], whose spectral-element code already does so; the mechanism is the one familiar from differential-difference equations. Only the measurements below are ours. Note also that second-generation points such as $L - \log 6$ exist, so the singular set is not limited to $|L - \log n|$ with n a prime power.

At $L = 1.19$ the Legendre coefficients of the lowest Galerkin eigenvector ($K = 384$) fall exponentially to about 10^{-25} by degree 160 and then only like $(\text{degree})^{-2.1}$. A periodogram of the high-degree coefficients locates singular points at $|y| \approx 0.088$–0.105, 0.183–0.198, 0.406, 0.510, 0.747, 0.884, 1.003, which match $|L - \log n| = 0.091$, 0.196, 0.419, 0.497, 0.756, 0.889, 1.007 for $n = 3, 4, 5, 2, 7, 8, 9$; there is no signal at $|L - \log 6| = 0.602$. A heuristic reason: in the eigen-equation the shifted copies $f(y \pm \log n)$ (with $f = 0$ outside $[-L, L]$) switch on or off where $y \pm \log n$ crosses $\mp L$, and only prime powers carry weight ($\Lambda(6) = 0$).

To quantify this we use a split-window basis (piecewise Legendre, discontinuous at $\pm(L - \log n)$; its x-quadrature error is checked by node doubling but not bounded) and record the one-sided jumps of the computed eigenvector at the breakpoints, normalised by $\Lambda(n) n^{-1/2} f(L)$. Table 9 gives the absolute values at $L = 1.19$ for four resolutions. The signs are negative for $n = 2, 3$ (where $L - \log n > 0$) and positive otherwise, as the heuristic predicts.

Table 9. $|\text{jump}|/(\Lambda(n) n^{-1/2} f(L))$ at $y = L - \log n$ (or $\log n - L$), $L = 1.19$, even sector, split-window basis with base allocation 24; "CV" is the coefficient of variation over the seven prime powers.

N_{tot}	$n = 3$	4	5	2	7	8	9	mean	CV
600	0.119	0.134	0.112	0.117	0.112	0.141	0.120	0.122	0.085
800	0.112	0.125	0.104	0.117	0.107	0.120	0.110	0.113	0.061
1200	0.107	0.109	0.103	0.107	0.104	0.119	0.118	0.110	0.054
1600	0.100	0.108	0.099	0.106	0.097	0.112	0.107	0.104	0.051

At $L = 1.19$ the normalised jumps of all seven prime powers are thus close to a common value (CV about 5% at the two highest resolutions), although raw weights $\Lambda(n) n^{-1/2}$ range from 0.25 to 0.74. The common value drifts down with resolution, so it is not a constant. As an effective mesh size we take $h = 2L/N_{\mathrm{tot}}$. A least-squares fit of $1/\text{mean}$ against $\ln(1/h)$, that is of mean $\approx c/(\ln(1/h) + b)$, to the four means (computed from the unrounded jumps)

gives $c \approx 0.740$, $b \approx 0.586$, with residuals below 0.0022; a power law $c'h^{0.153}$ fits the four means equally well, so they do not determine the law. The data are *consistent with* the computed jumps being finite-resolution images of a logarithmically softened singularity of a continuous function, and the jumps *appear* to tend to 0; four resolutions do not establish either statement, nor anything about the limiting eigenfunction. The raw measurements (one record per resolution, with every jump, weight and $f(L)$) are `data/kinks/kink_119_<N>.json`. At $L = 1.5$ the value-jump ratios are again close to each other, but the split-window mesh there omits some cuts (including second-generation points close to the $n = 8, 16$ cuts), and we make no per-breakpoint claim. Prior work: the location of the singular points and the cut mesh are Eberwein's [14, §13]; jumps of a trial vector at $L - \log n$ appear in Mori's work [26], and the $\pm \log n$ shift structure in [33, eq. (2.11)].

10. Reproducibility

10.1. **Software and hardware.** Python 3.12 (3.12.13 for Pieces I–II) with python-flint 0.9.0, which bundles FLINT/Arb 3.6.0 [34, 20]; the C tools `vchol` (verified Cholesky) and `veig` (estimates only) of Piece II are built with gcc 15.2.0 against FLINT 3.4. Pieces I–II ran on a 20-thread laptop and a cluster of three 64-core nodes (FLINT versions on the cluster not recorded; a SHA-256 check of the code files on every host was in force for the certificate of record). Piece III and §8 ran on three 64-vCPU virtual machines. Elston's certifier is used at upstream commit `3cc4cb7be6a659f1463ae4e2e99ad7ae5e6646f5` with the `--deflate` option added (`code/gap/elston/`). Piece IV ran on one 64-vCPU virtual machine with 60 GB of memory, with Python 3.12.15, python-flint 0.9.0, mpmath 1.4.1 and numpy 2.5.4, and a C kernel for Lemma 7.4 that gate P0 checks against the Python reference; the hashes of all executed files are in the run-time stamp and in every run record.

Run times. Piece II certificate of record: clean rerun for $L \le 1$ about 5.8 h in three stages; extension 1.03–1.09: 2859 s, 11022 s, 11760 s; continuation 1.12–1.19 about 28.5 h in total (head assembly about 6.5–7 h per window on the cluster). Piece III test at $L = 1.19$: 10–12 min per run with 18 workers. Piece III certificates: per run as in Table 3; the 16 runs behind Theorem 4.1 took about 814 worker-process hours (seconds times workers), all 24 shipped runs about 2006. Upper bounds U ($K = 640$, 3000 bits): 626–822 s per window. Rigorous upper bounds of Table 8: $L = 2$, $K = 1024$: 4276 s (even, rerun; the original even run took 3166 s), 10971 s (odd); $L = 2.25$: 63923 s (even), 77675 s (odd); $L = 1.5$, 1.75 odd: 1278 s, 1293 s; $L = 1.5$ even ($K = 384$): 421 s.

10.2. **Commands.**

- Piece I and Piece II gluing: `python code/theorem.py --check` (seconds; ends with "THEOREM CERTIFIED for all L in (0, 1.19]"). Windows: `python code/theorem.py` with the environment settings recorded in the release.
- Piece III gluing: `python code/gap/check_steps.py` (seconds; reads `data/piece3/`; ends with "CERTIFIED: …simple-even for all L in (0, 1.4885]", compare `data/piece3/check.log`; only $(0, 1.4035]$ is used).
- Piece III lower bounds, in `code/gap/elston/`, with the window options `<w>` from the field `args` of the certificate file (e.g. `--a 1.2205 --delta 0.3 --beta-k 64 --tsharp 850 --nmodes 849 --gl 56 --prec 400`):

```
C="--panels-per-unit 4 --schur-k 4000 --schur-iters 1500"
python grid_certificate_rigorous.py <w> $C --sector odd --workers n --json <out>
python grid_certificate_rigorous.py <w> $C --sector even --deflate --workers n \
       --json <out>
```

- Piece IV step test: `sh check_theorem.sh` in the repository root runs the gluing of Pieces I–IV on the shipped data (seconds). Tables 6, 7 and the generated rows of Tables 3 and 4 (seconds; the script checks every glued record and its gates and every step in exact arithmetic on the Arb endpoints, and optionally, given the full run directories, that each glued record is the registered selection):

```
python paper/gen_tables.py data/piece4/glue data/piece4/windows.txt \
       data/piece3 > paper/generated.tex
```

  The command line of each Piece IV certifier run is in the field `argv` of its record; `code/rh60route/README.md` explains how to re-run a window.
- Upper bounds, in `code/upper/` (exports the outward-rounded field `lam0_upper`):

```
THREADS=n python fr_sweep.py <out.jsonl> "640" "1.19,1.2205,..." 3000
```

10.3. **Verification and audits.** All reviews so far were carried out by AI agents directed by the author; no independent human referee has checked the proofs or the code.

- *Pieces I–II.* Three independent review rounds of the earlier note produced, among others: the Gauss–Legendre exponent correction (Remark 5.9); reconciliation of the Piece I constants across precisions; outward rounding of all quoted bounds; a downward-induction proof of the Bessel ratio enclosure; the definition of the continuation of $\operatorname{Re}\psi$ off the real axis; documentation of Python `min` on balls; a mollification argument for (2.2); a self-contained discreteness proof; a SHA-256 check of remote code; and corrected credit to Zhu for the reduction. Every finding was checked against code, logs and data before it was applied. A clean end-to-end rerun with a single code revision (empty cache, tools rebuilt, remote code hashed) then reproduced all lower bounds for $L \le 1$ to about three significant digits. The sharp-cutoff path and the windows 1.03–1.19 have not been through a review round.
- *Upper-bound code.* An external review of the high-precision notes (six findings, all accepted) led to per-code-path rigour labels, outward rounding of exported upper endpoints, and withdrawal of an unsupported tail claim; an earlier review found and fixed an inaccurate ball inverse, a missing residual check and a missing quadrature bound.
- *Piece III.* Elston's certifier was audited independently at $L = 1.19$ (two issues fixed; documented in [16]); the post-audit code at larger L and our `--deflate` addition have not been audited. The step test (`code/gap/check_steps.py`, about 50 lines) compares explicit Arb endpoints. The reproduction of Elston's $L = 1.19$ bounds to 12 and 17 digits is a consistency check of the installation, not an audit.
- *Piece IV.* The specification and the proof document were reviewed by a fresh AI agent before the pilot was pre-registered. A pilot at $L = 2$ was audited by another fresh AI agent with its own code (matrix entries from (2.2), far-block constants, an independent core solve, mutation tests, rounding directions, clean room); it found no rigour defect and required six fixes (independent per-window entries, an uncapped second path, explicit labels for Lemma 7.4, archiving of executed code, an append-only amendment log, checks at gluing), which were made and verified before the grid was pre-registered. The grid was then checked by a fresh AI agent with its own code (step test, gates, controls, mutations, provenance, entries from (2.2), the cross-check), and the gap closure and the combined statement on $(0, 2]$ by another fresh AI agent, whose verdict is "certified, conditional on the imported trust of §7.9; grid G alone gives $[1.4035, 1.91912841796875]$". None of these agents wrote the code it checked, but all are instances of the same AI system that wrote it.
- *Second AI system.* OpenAI's Codex reviewed the release candidate of this version on 1 November 2026, read-only and with web search. It re-derived the parity, monotonicity,

TABLE 10. Provenance of the numbers in this paper. Paths under `highprec/` are in the author's working tree and are not part of the repository; all files the theorem depends on are. Shipped run records that contained local paths or host names are redacted copies; a list mapping the SHA-256 of each original to that of its copy is shipped with them, and the originals are available on request.

Item	File(s)
Table 2, Piece I constants	`data/theorem_bounds.json`; `code/theorem.py`; logs `data/logs/check.log`, `run_rerun.log`, `run_extend.log` (release of [5])
Remark 5.9	`data/history/theorem_bounds_corrected.json`, `data/logs/run_gl_correction.log`
Piece III parameters (Table 3)	field `args` of each file in `data/piece3/`; design rule `code/gap/params.py`
Piece III certifier	`code/gap/elston/grid_certificate_rigorous.py`, `UPSTREAM_COMMIT`, `LICENSE`
$L = 1.19$ test (§6.2)	run logs on the compute host, not shipped (the test enters no step of the proof)
Table 4 lower bounds	`data/piece3/g_<L>_l2.json` (λ_2^{ev}), `g_<L>_odd.json` (λ_1^{odd}); field `certified_lower_bound`
Table 4 upper bounds	`data/piece3/upper640.jsonl`, field `lam0_upper`; code `code/upper/`
Piece III gluing	`code/gap/check_steps.py`, log `data/piece3/check.log`, checksums `data/piece3/SHA256SUMS`
Piece III past 1.4035 (Tables 3, 4 below the rule; Table 7)	`data/piece3/` (not needed for the theorem)
Table 6	glued records and upper bounds `data/piece4/glue/`, window list `data/piece4/windows.txt`; fields `certified_lower_bound`, `lam0_upper`, `gates`; generator `paper/gen_tables.py`, output `paper/generated.tex`
§7.6	the four λ_2^{ev} records at 1.92364501953125 in `data/piece4/rule_e/`; controls `data/piece4/controls/`; gluing logs `data/piece4/logs/`
Piece IV protocol	pre-registrations, stamps, amendment log, scored results and audit in `data/piece4/`
Piece IV certifier	`code/rh60route/`; specification, proofs P1–P23 and the formula–code–field map in `supplement/` (`PROOFS.md` byte-identical to the stamped file; `SPEC.md` redacted, `REDACTION.sha256`)
Table 8, $L = 2$	`data/piece4/glue/upper_rerun_L2.jsonl` (both sectors, rerun of 31 October); superseded original even run `data/upper/rig_L2.jsonl`
Table 8, other rows	`data/upper/rig_L225.jsonl` ($L = 2.25$ even), `data/upper/odd.jsonl` (odd sector, $L = 1.5$–2.25), `data/piece4/glue/upper_k384.jsonl` ($L = 1.5$ even)
Non-rigorous Galerkin values (§8)	`highprec/data/fr_L225*.jsonl`, `fr_sweep*.jsonl`
§9, Table 9	`data/kinks/kink_119_<N>.json` (field `breaks`: `jump_f`, `weight`; and `f_at_L`); $L = 1.5$: `highprec/data/kink_15_*.json`, `tail_probe.json`

deflation and Schur-complement arguments, recomputed the Piece I constants, ran the gluing script, rebuilt `paper/generated.tex`, and compared 40 Legendre–Galerkin and 18 trigonometric matrix entries with independent high-precision code (agreement to

$6.5 \cdot 10^{-62}$ and $1.4 \cdot 10^{-55}$). It found no fatal error and no counterexample to the theorem, but recommended against release as a completed computer-assisted proof until three major points were fixed: the Piece IV derivations were not public; two printed upper bounds of Table 8 lacked a demonstrated outward rounding; and the hypothesis of [13, Thm. 6.1] was misstated. It also raised nine minor points, among them that rule E is not needed (Remark 7.5). This version addresses all twelve; the review and our point-by-point responses are in `reviews/`. Codex did not regenerate the window matrices, stage S, the C factorisations or the multi-hour certificates, and it did not certify the proof.

Invitation to review. This work has not been peer-reviewed. Code, certificate files and this source are public at `https://github.com/AlexGenusCore/weil-ground-state-simple-even`; the file `REVIEWING.md` there lists the least-checked parts and concrete checks ordered by effort, from re-running the gluing script on the shipped certificates (minutes) to re-running a single window (hours). Independent review by anyone is welcome; confirmed errors will be corrected publicly and reviewers credited.

11. Limitations

(1) *Range.* The theorem is certified on $(0, 2]$, conditional on the imported trust of §7.9.
(2) *Post-hoc rule in Piece IV.* The pre-registered grid failed as registered (§7.6). The step $[1.91912841796875, 1.92364501953125]$ that it left open is closed by rule E, which was registered after the failure and after a diagnostic run had passed and changes only the free parameter N. It is also closed without rule E, by Lemma 3.1 from a registered record at the next window (Remark 7.5); that substitution was likewise written down after the failure, but uses no new computation. Rule E was tested at one window only, and its negative control failed at (V2) rather than at (V4), so it exercised the Lemma SR path at the new N less than planned.
(3) *Origin of the Piece IV method.* The method comes from AI-agent preprints accepted only by AI reviewer agents [23, 8]. We re-derived every step, but no human has checked our derivation or our code.
(4) *Stage S.* An upper bound $s(L)$ for the prime comb that is too small would make Piece IV certificates too strong and would not be caught by any gate except the coarse Ritz guard G-S; $s(L)$ is computed with Elston's Schur-test code, audited only at $L = 1.19$.
(5) *Piece III is unaudited beyond $L = 1.19$.* Elston's certifier was audited only at $L = 1.19$; the runs at larger L, the parameter changes (β_K, $T^\sharp$, N, Gauss points) and our `--deflate` addition have not been reviewed. The step ratios of Piece III are small (1.131–1.552), so the theorem on $(1.19, 1.4035]$ depends on every bound being correct to within about 13%. The step ratios of Piece IV are at least 2.098.
(6) *Computer assistance.* Correctness depends on Arb/FLINT, on our code for Pieces I–II, Piece IV and §8, on Elston's code for Piece III and for parts of Piece IV, and on the lemmas above.
(7) *Normalisation.* Our Q_L follows Zhu and was validated against the zero side of the explicit formula. The second AI review matched its pole, prime and archimedean terms against [12, (3.7)–(3.11)] (§1); no human has checked this line by line. Remark 2.4 covers positive affine changes, not a discrepancy in the relative weights of the pole, archimedean and prime terms.
(8) *Self-adjoint realisation.* We use the closure of Q_L on $C_c^\infty(-L, L)$ (§2); trial polynomials not vanishing at $\pm L$ are in its form domain by the argument in §5.2. We have not shown that trigonometric polynomials form an operator core for A_L, which is the hypothesis

of [13, Thm. 6.1]; so no statement about the zeros of the Fourier transform of the ground state follows from our theorem as proved here. A different realisation would need re-examination.

(9) *Unexplained precision dependence* of the upper end of $\int |h|$ in Piece I (largest value used).

(10) *Comparisons on balls.* Python `min`/`max` on Arb balls occur in the gluing and in some intermediate steps of both certifiers; every current use is separated by many orders of magnitude more than the radii.

(11) *Sharp-cutoff and triangular assembly code* for $1.03 \le L \le 1.19$ in Piece II has not been reviewed; $L = 1.19$ is, however, re-confirmed by the Piece III code path.

(12) *Upper bounds of §8* are upper bounds only; their distance from the true eigenvalue is not controlled (Remark 8.2 for the quadrature exponent at $L = 1.5$ and $L = 2$).

(13) *Section 9* is numerical; the split-window x-quadrature is not bounded.

(14) *Scope.* Nothing is claimed for $L > 2$, about the convergence step (ii) of [12], or about RH.

Acknowledgements

The Piece III lower bounds use Zac Elston's certifier *weil-windows* [16] (MIT licence), which we ran with one added option; the reduction it implements and the published $L = 1.19$ certificates are his. The window certificates of Pieces I–II rest on Xuefeng Zhu's frequency-cutoff reduction and block estimate [38]. The survey of lower-bound methods behind Piece III drew on the work of Mori [26] and of Liu [24], whose position-side use of the Legendre convolution recurrence preceded ours. The method of Piece IV is that of bramble_trace [8] and lihongmin [23] on rStation; we re-derived and re-implemented it without using their code.

AI-assistance statement

Computations, code, the Piece IV specification and proof document, and drafting were carried out with substantial assistance from an AI system (Anthropic's Claude). All reviews and audits so far were carried out by AI agents (§10.3); no human referee has checked the proofs or the code. A second AI system (OpenAI's Codex) reviewed a release candidate (§10.3). The main results were re-checked by independent code as described in §10.3, within the limits of §11, and the author takes responsibility for the content. The method of Piece IV comes from AI-agent preprints [23, 8] that were reviewed only by AI agents.

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